

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

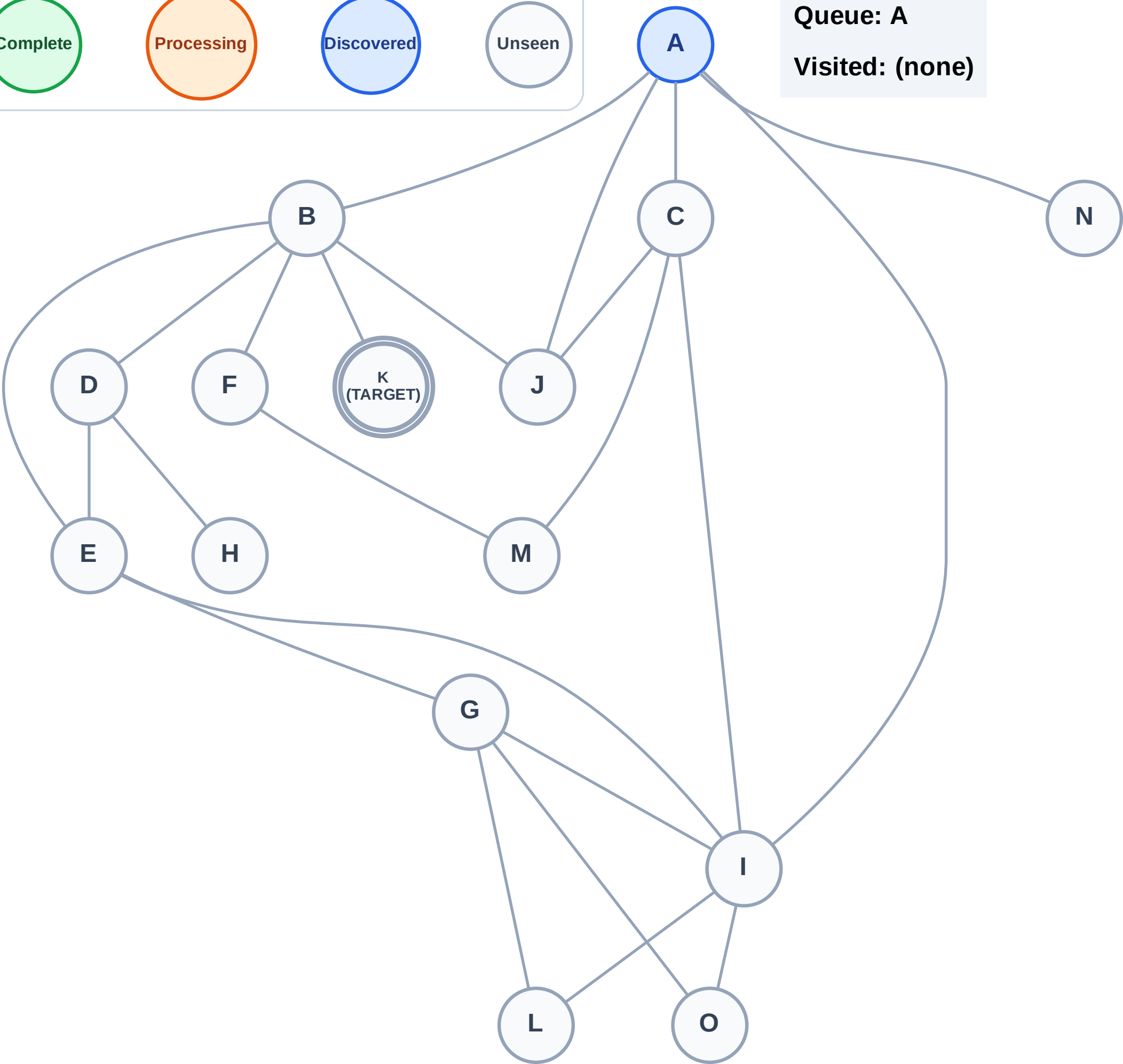
Complete

Processing

Discovered

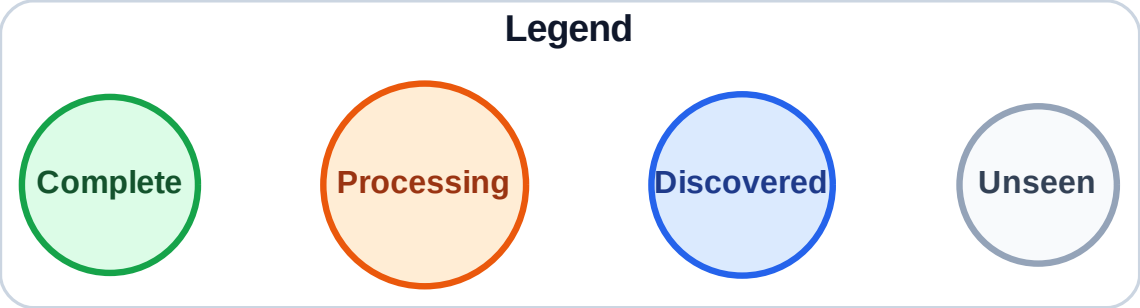
Unseen

Queue: A
Visited: (none)



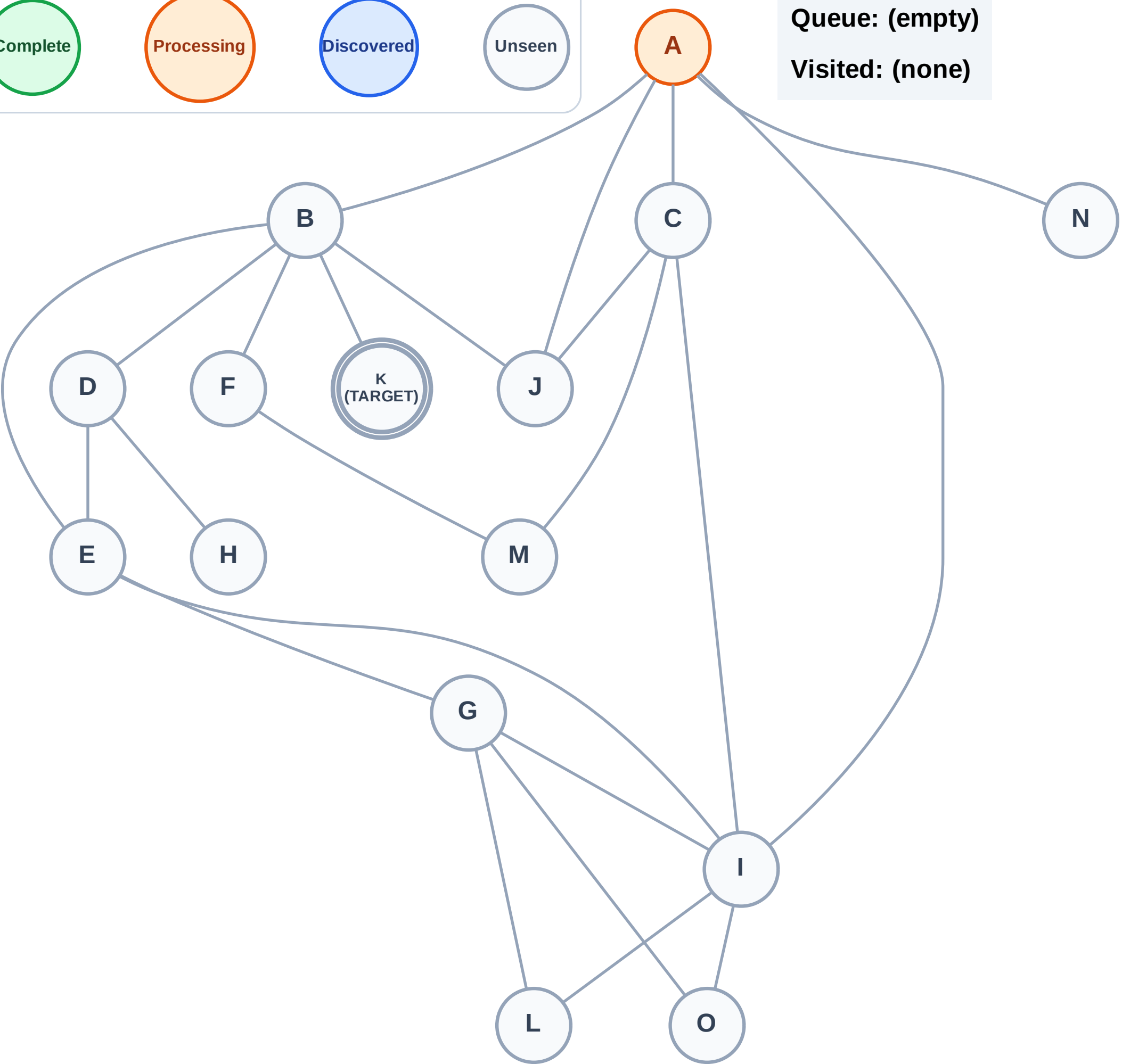
BFS Traversal

Step 2: Process node A



Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, C, I, J, N from A

Legend

Complete

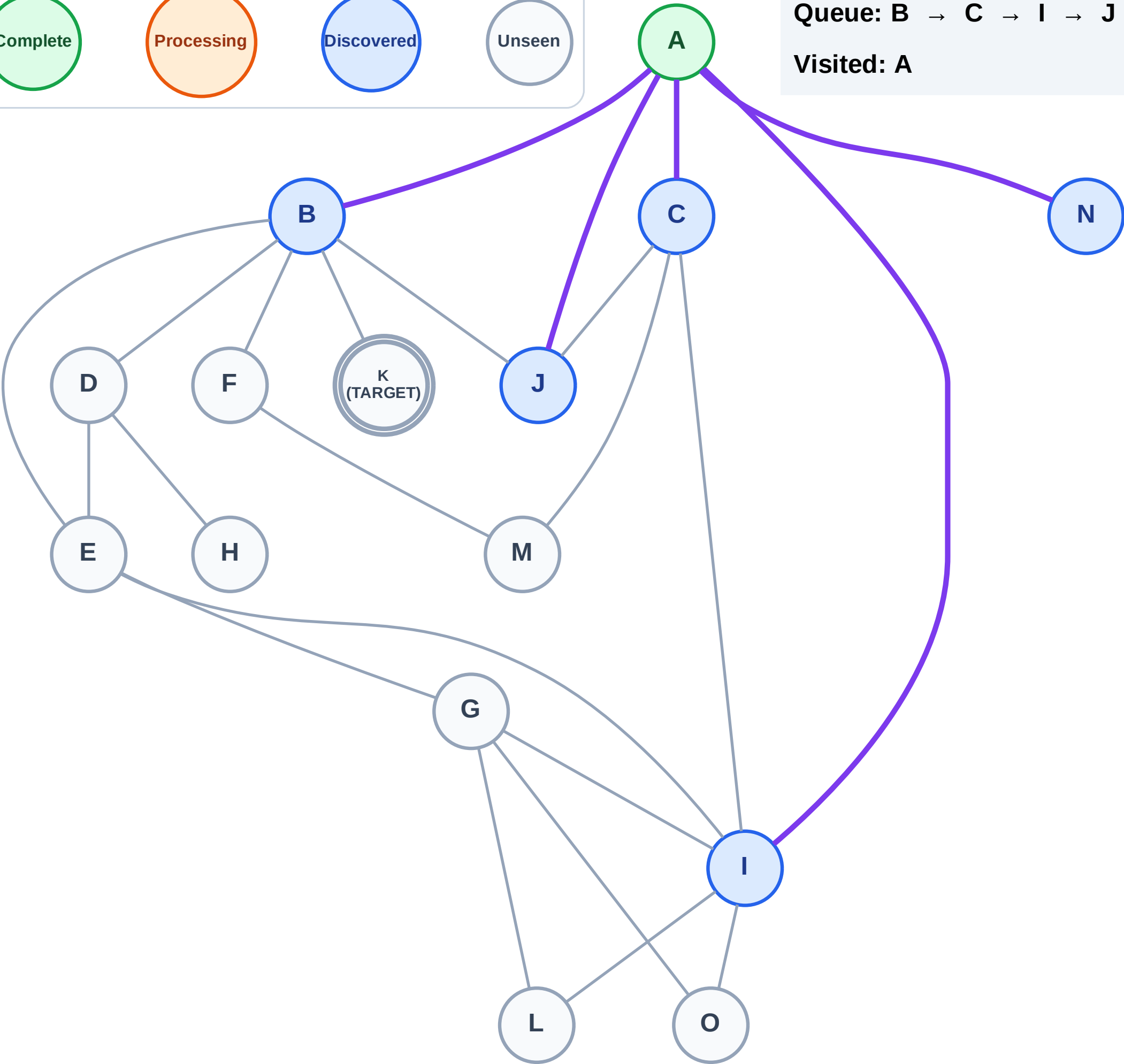
Processing

Discovered

Unseen

Queue: B → C → I → J → N

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

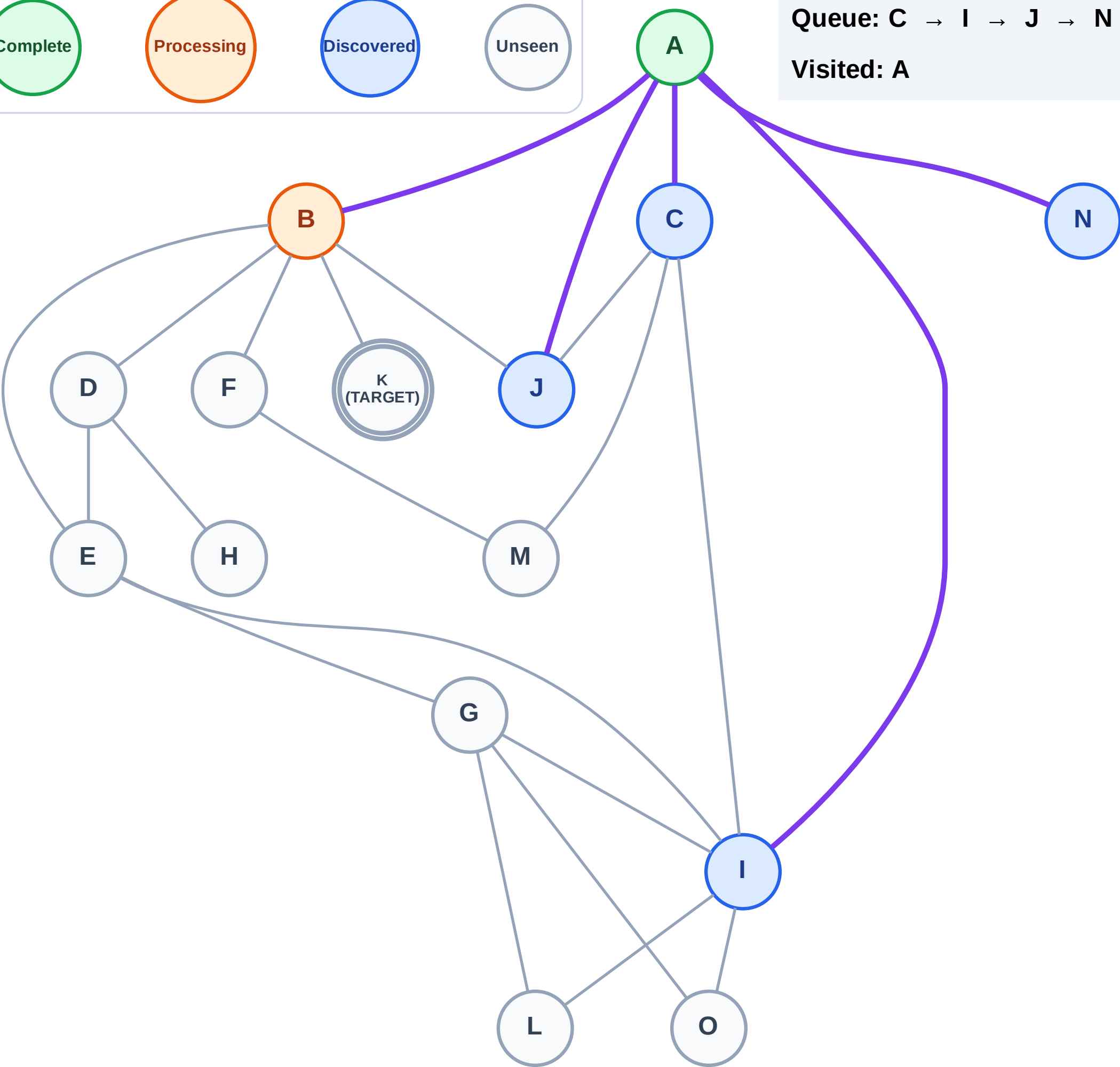
Processing

Discovered

Unseen

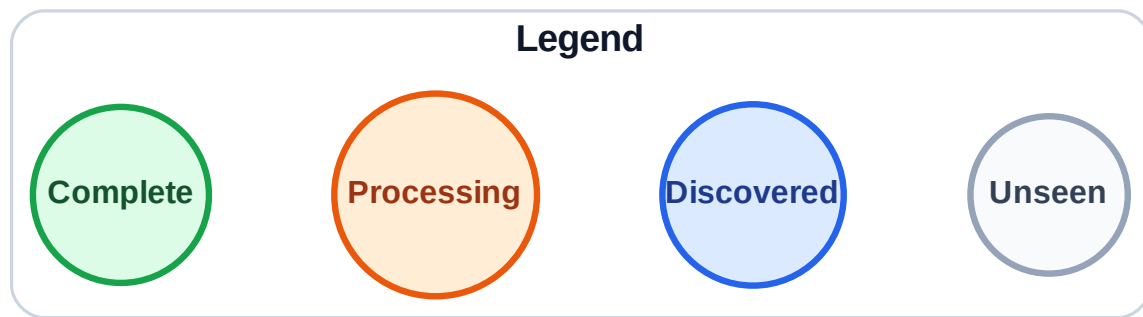
Queue: C → I → J → N

Visited: A

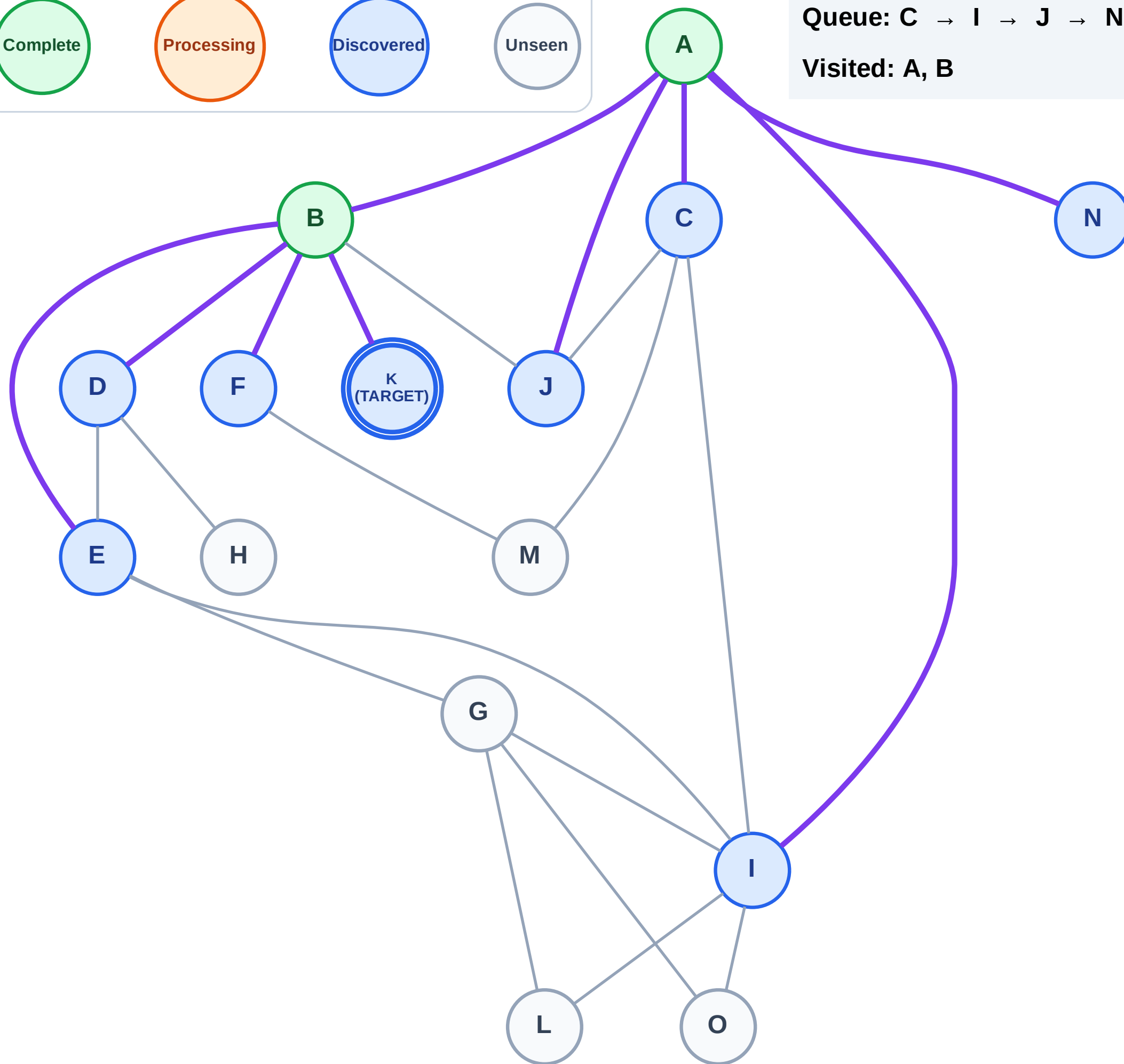


BFS Traversal

Step 5: Discover D, E, F, K from B



Queue: C → I → J → N → D → E → F → K
Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

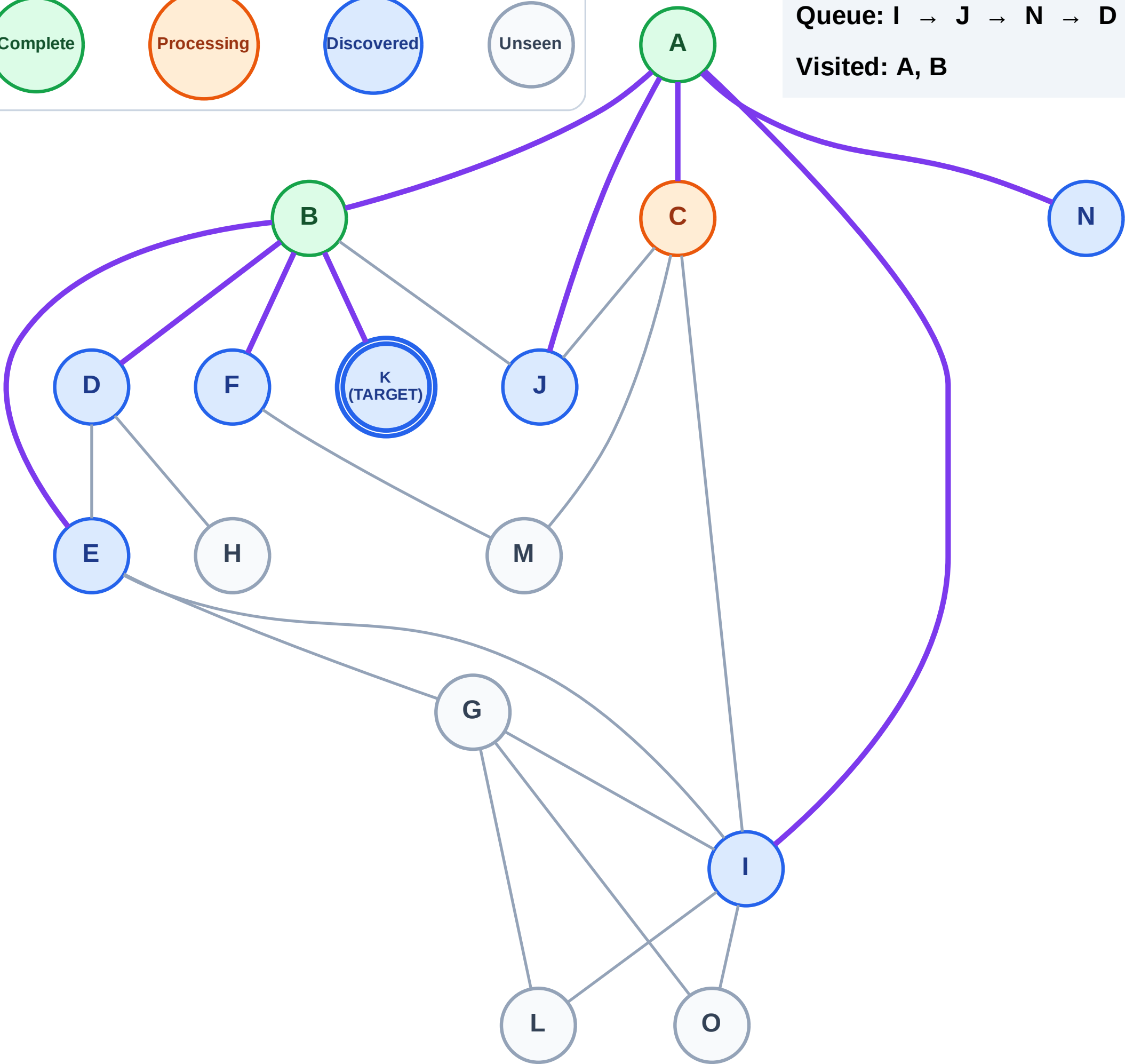
Processing

Discovered

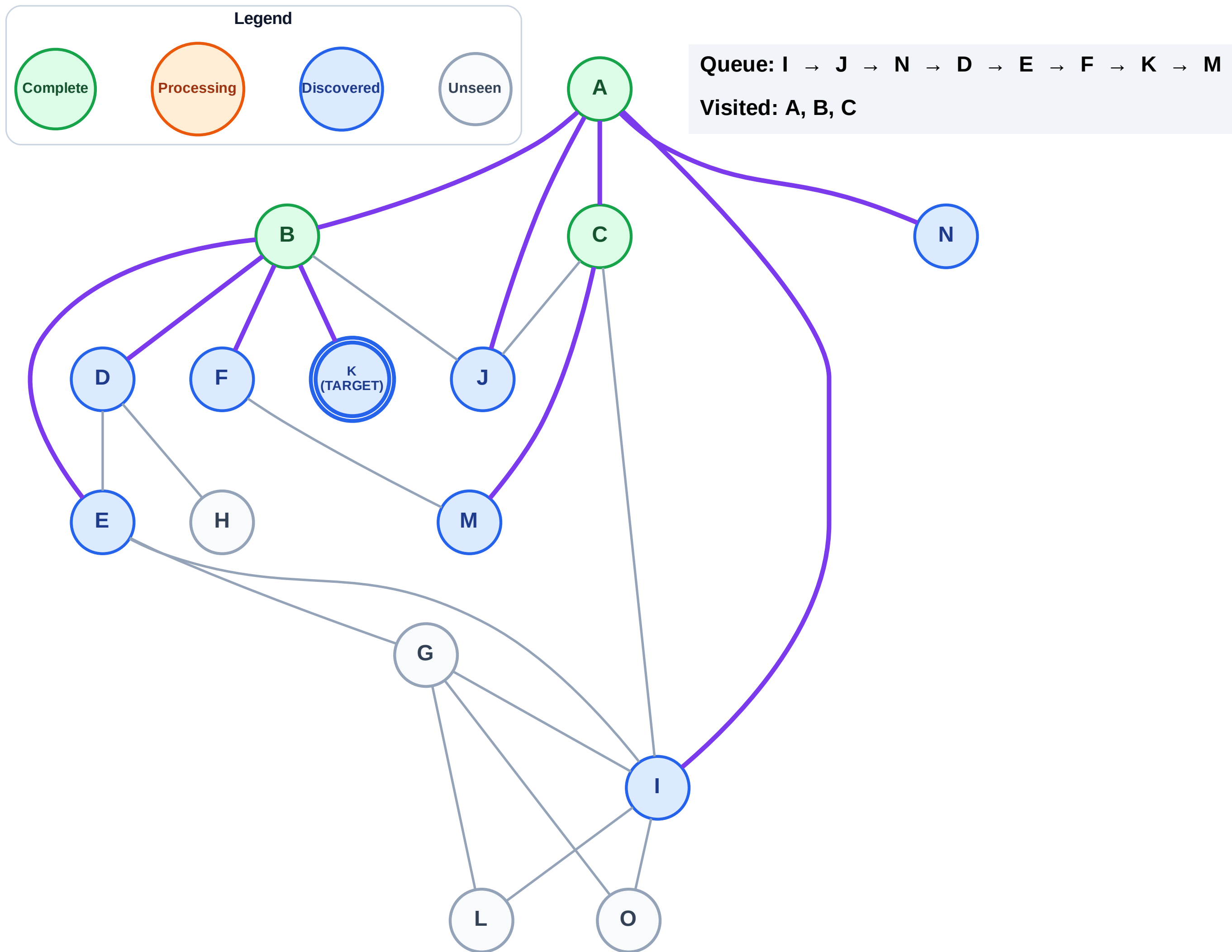
Unseen

Queue: I → J → N → D → E → F → K

Visited: A, B



Step 7: Discover M from C



BFS Traversal

Step 8: Process node I

Legend

Complete

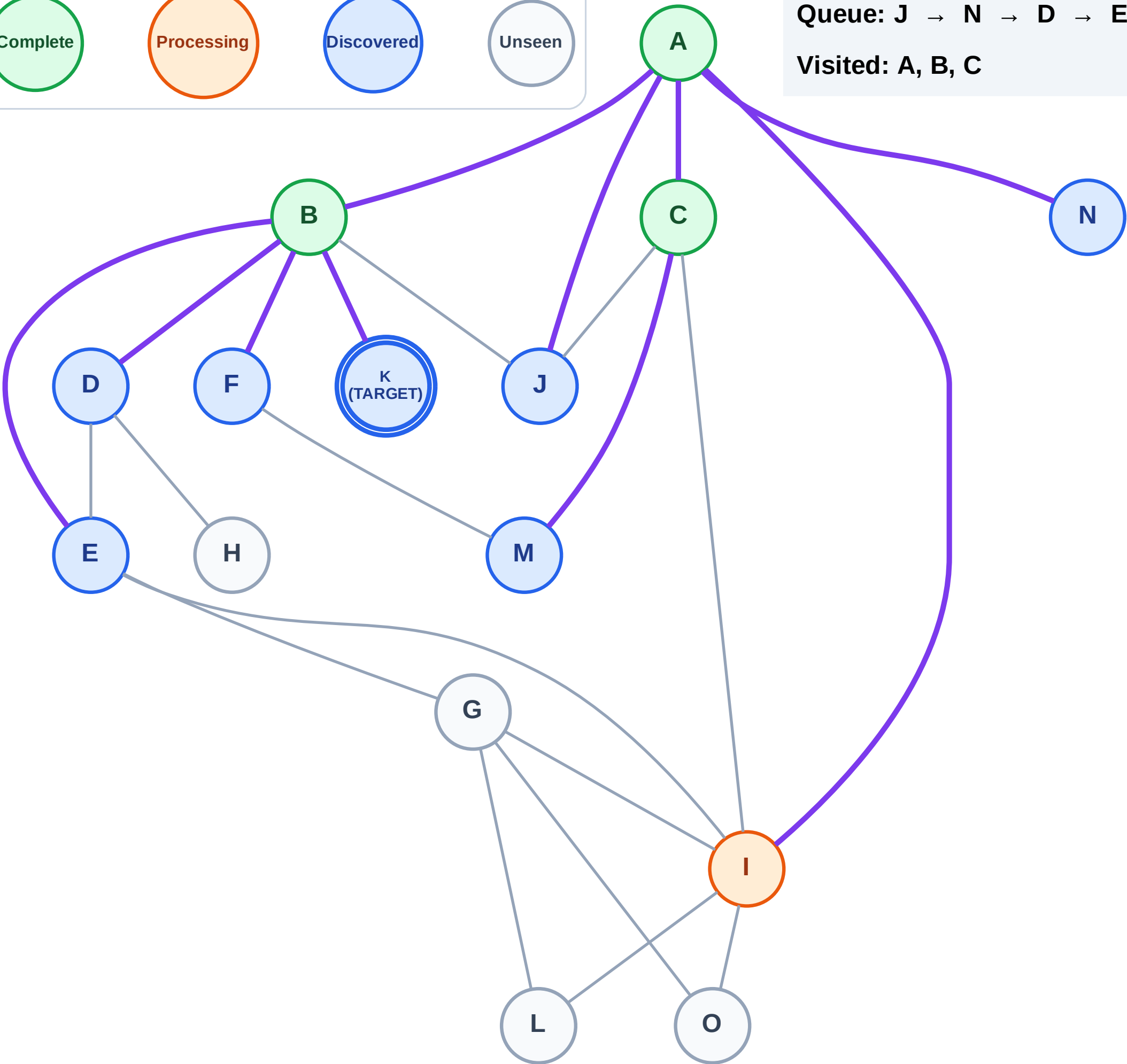
Processing

Discovered

Unseen

Queue: J → N → D → E → F → K → M

Visited: A, B, C



BFS Traversal

Step 9: Discover G, L, O from I

Legend

Complete

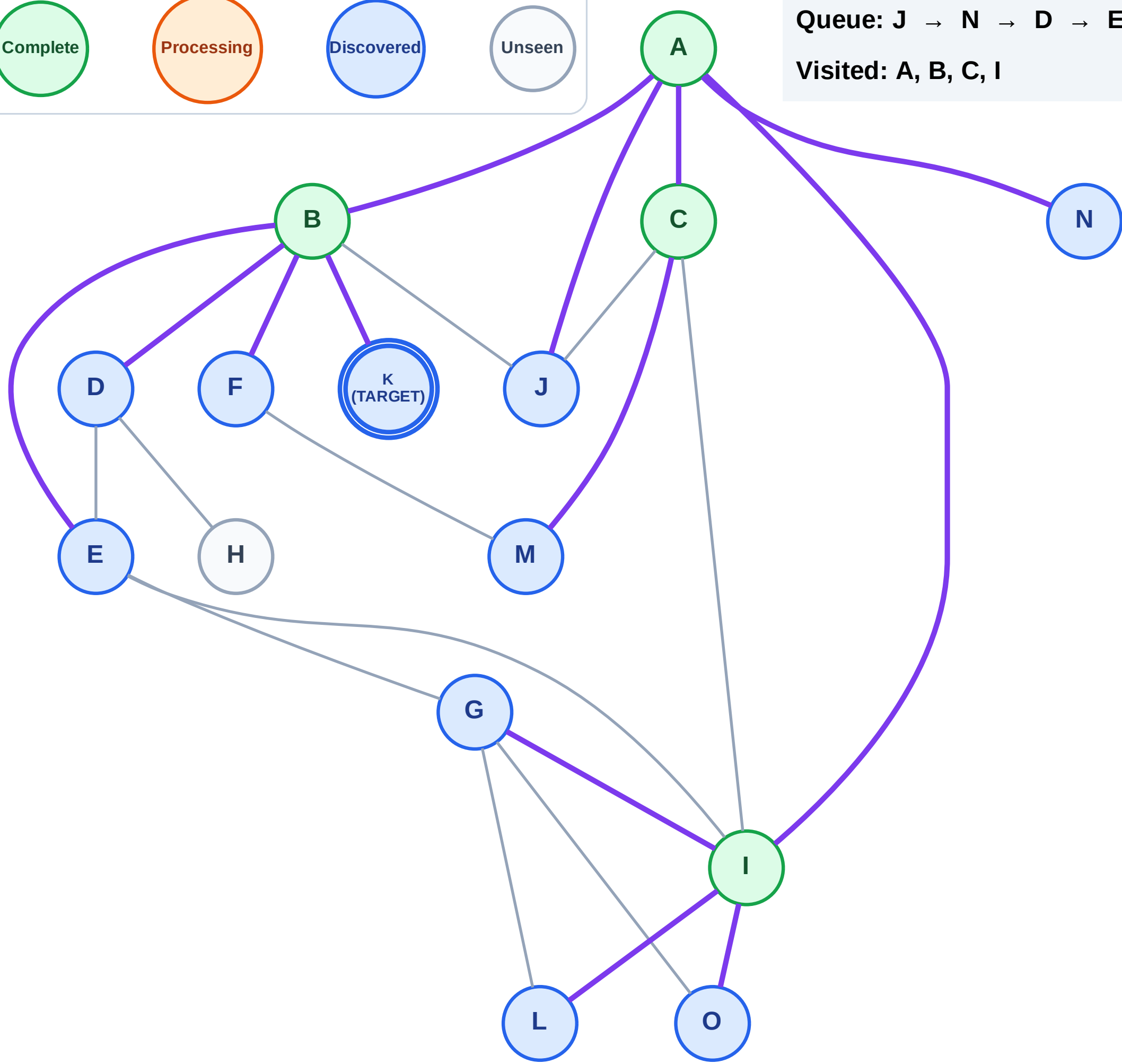
Processing

Discovered

Unseen

Queue: J → N → D → E → F → K → M → G → L → O

Visited: A, B, C, I



BFS Traversal

Step 10: Process node J

Legend

Complete

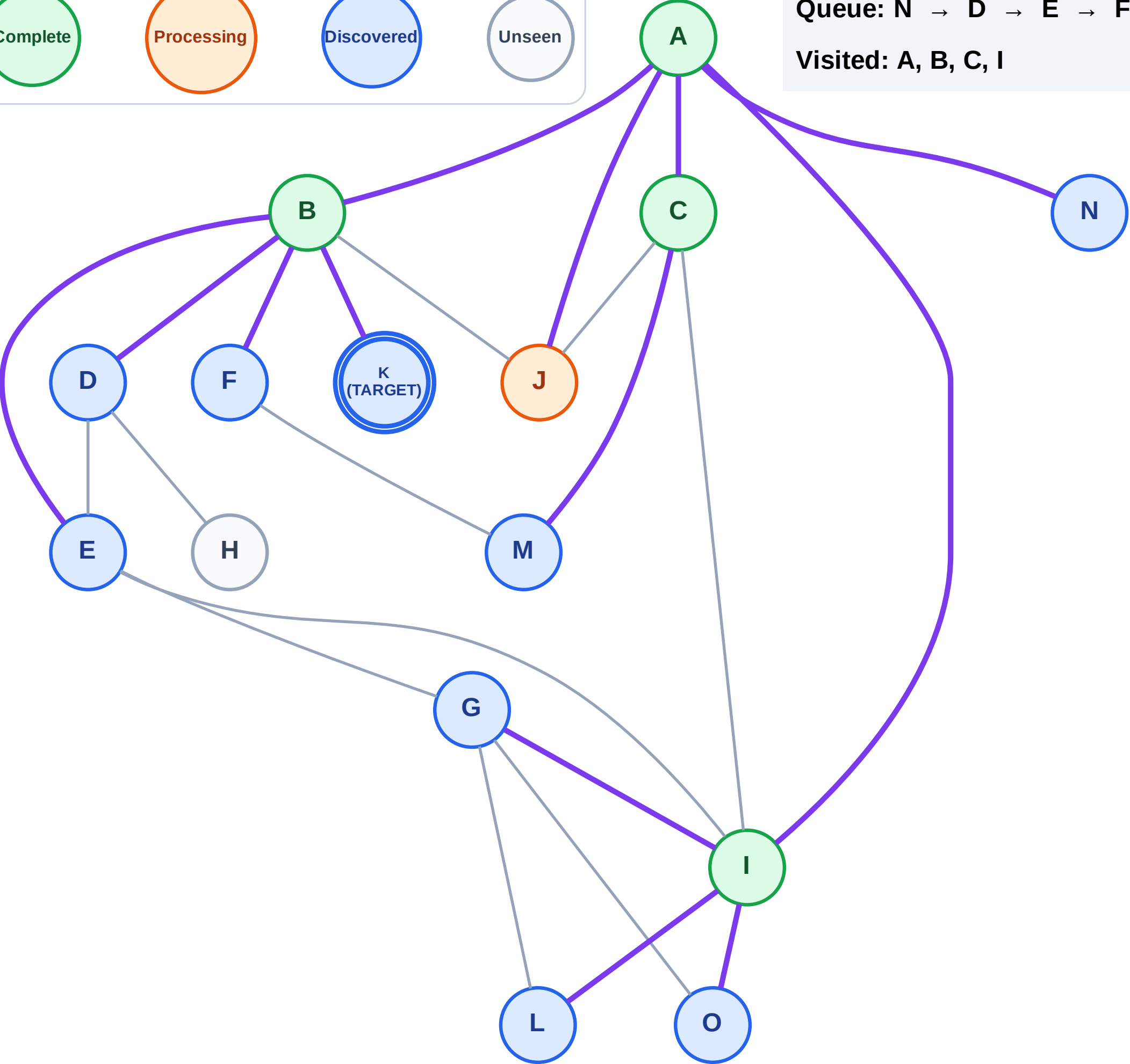
Processing

Discovered

Unseen

Queue: N → D → E → F → K → M → G → L → O

Visited: A, B, C, I



BFS Traversal

Step 11: No new neighbors from J

Legend

Complete

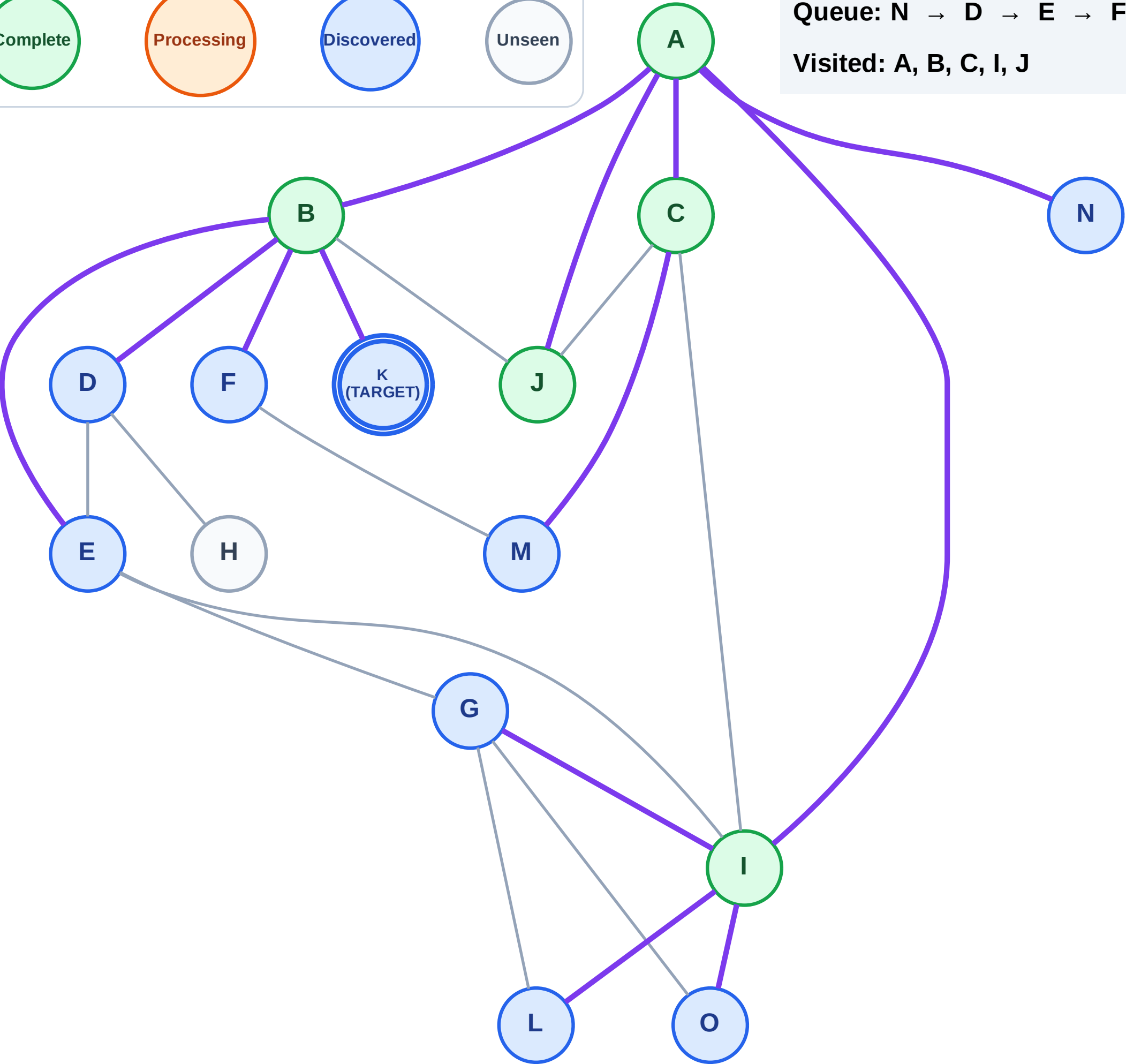
Processing

Discovered

Unseen

Queue: N → D → E → F → K → M → G → L → O

Visited: A, B, C, I, J



BFS Traversal

Step 12: Process node N

Legend

Complete

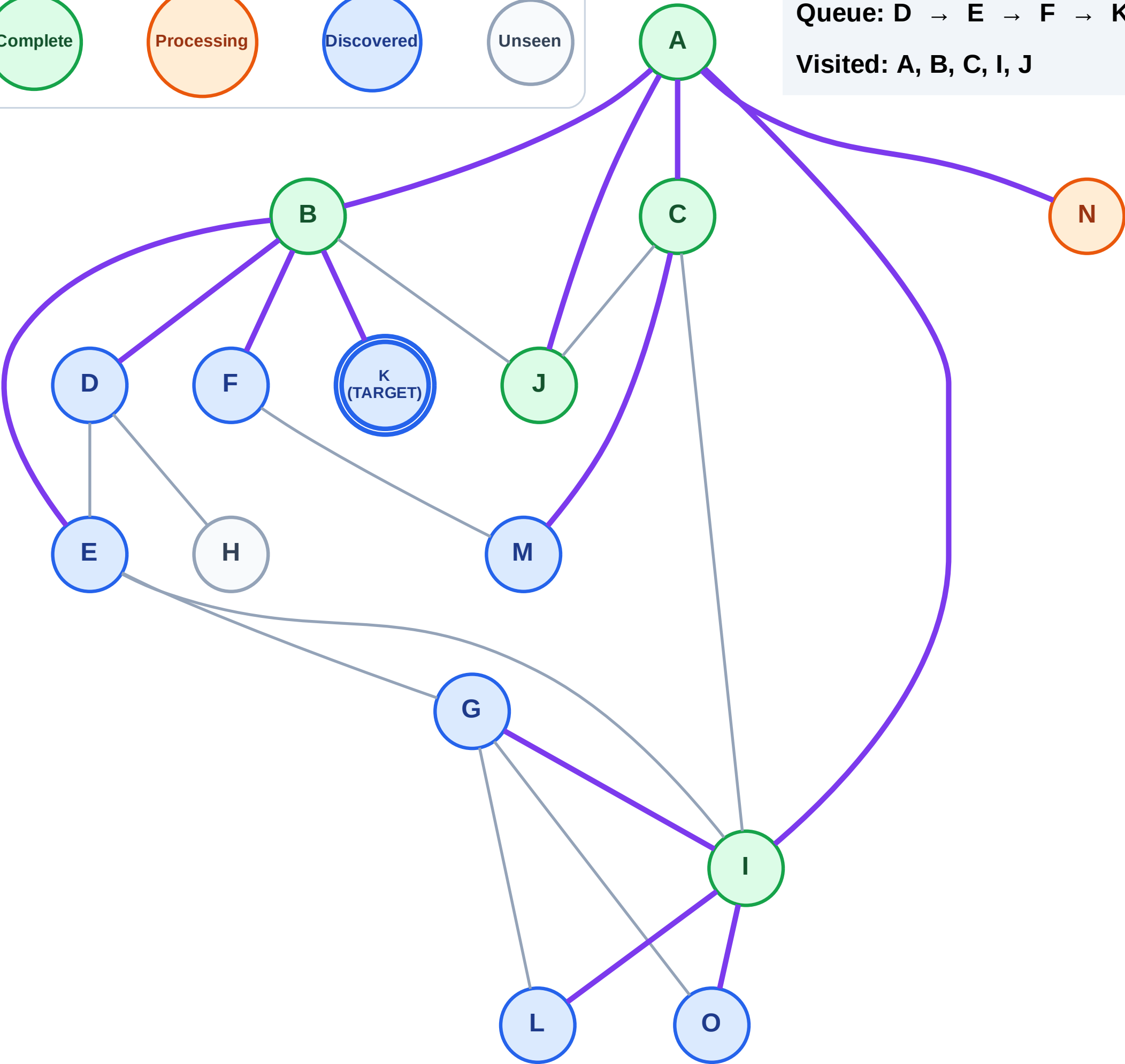
Processing

Discovered

Unseen

Queue: D → E → F → K → M → G → L → O

Visited: A, B, C, I, J



BFS Traversal

Step 13: No new neighbors from N

Legend

Complete

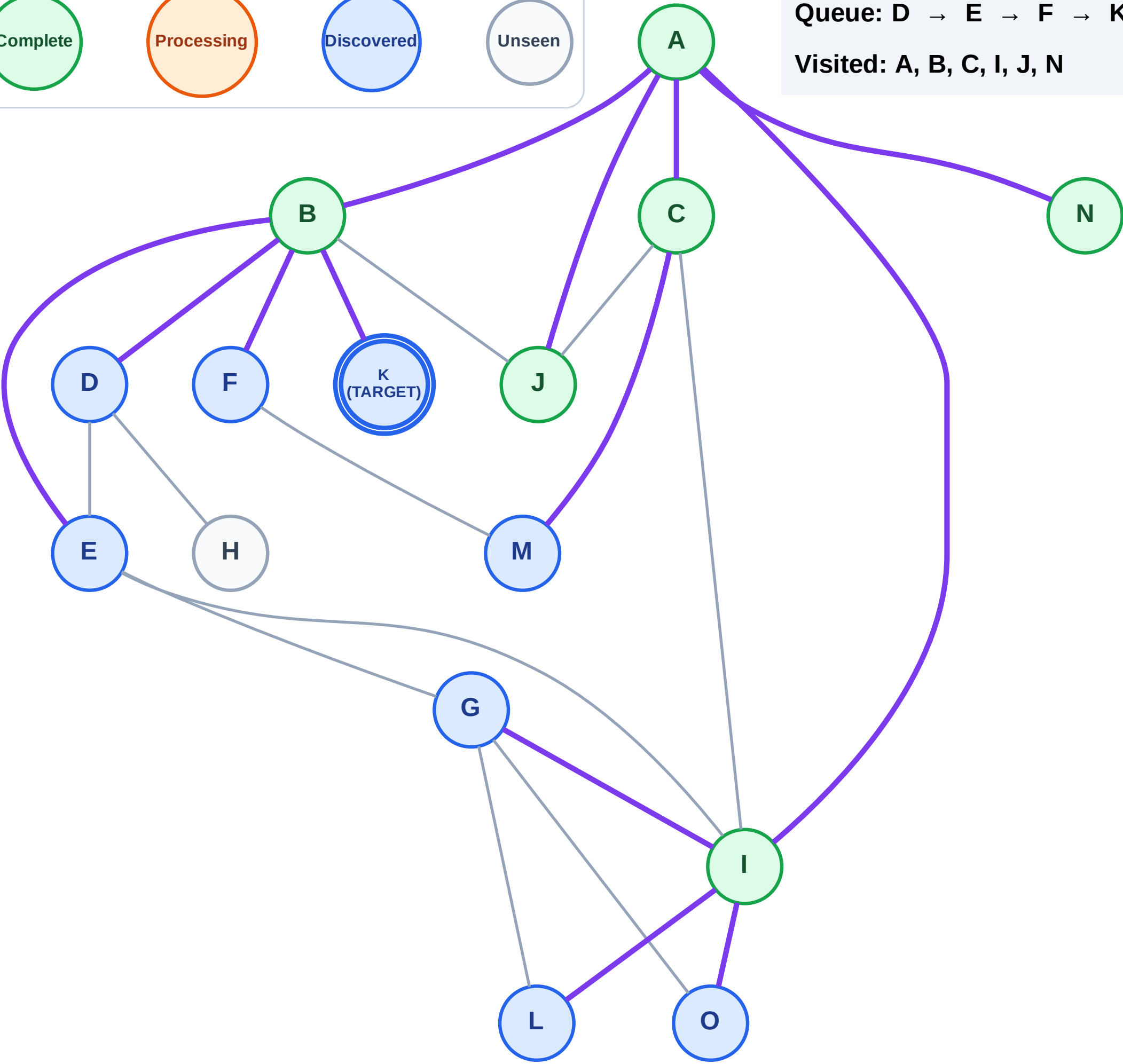
Processing

Discovered

Unseen

Queue: D → E → F → K → M → G → L → O

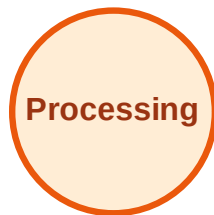
Visited: A, B, C, I, J, N



BFS Traversal

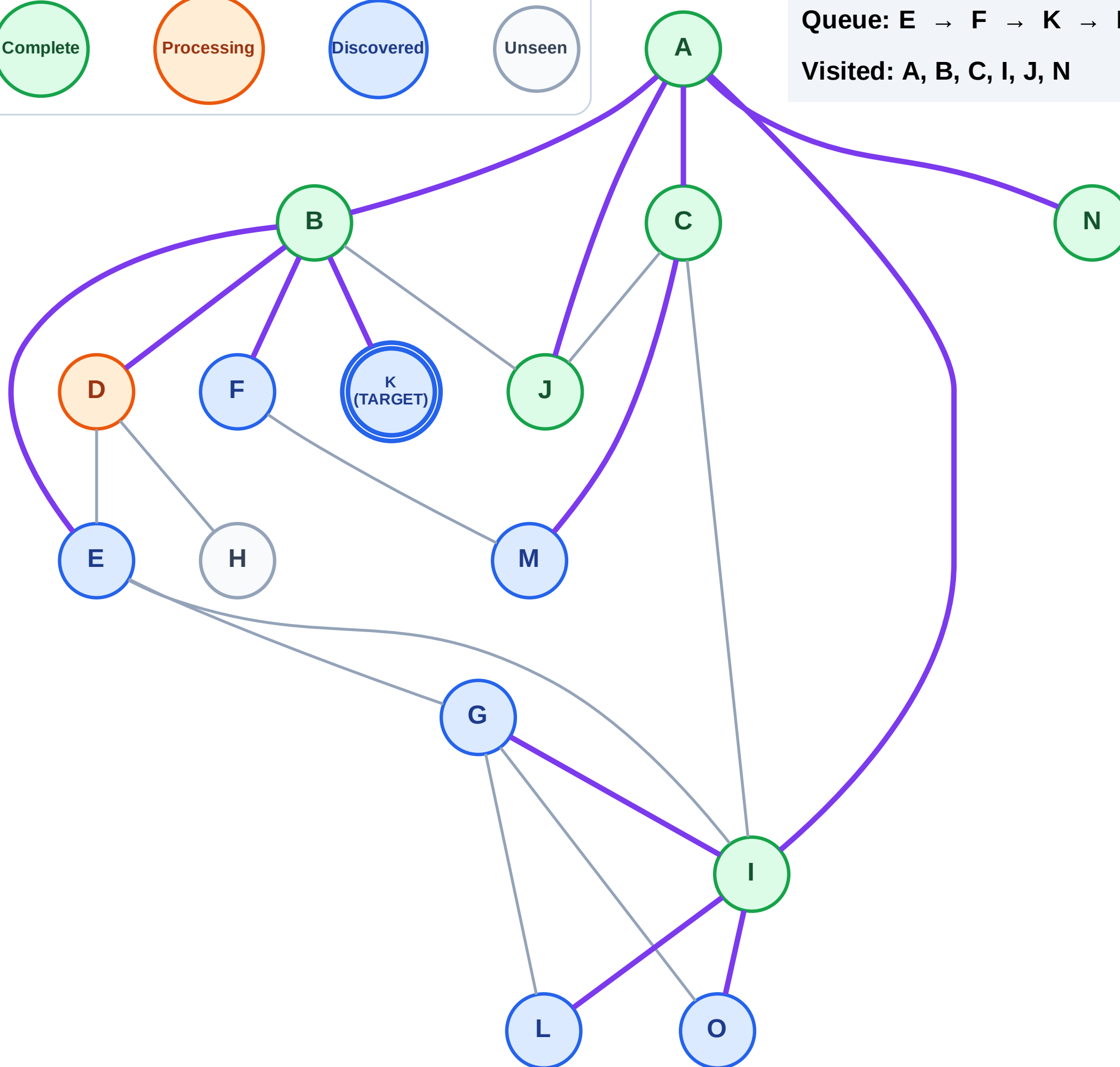
Step 14: Process node D

Legend



Queue: E → F → K → M → G → L → O

Visited: A, B, C, I, J, N



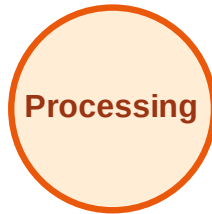
BFS Traversal

Step 15: Discover H from D

Legend



Complete



Processing



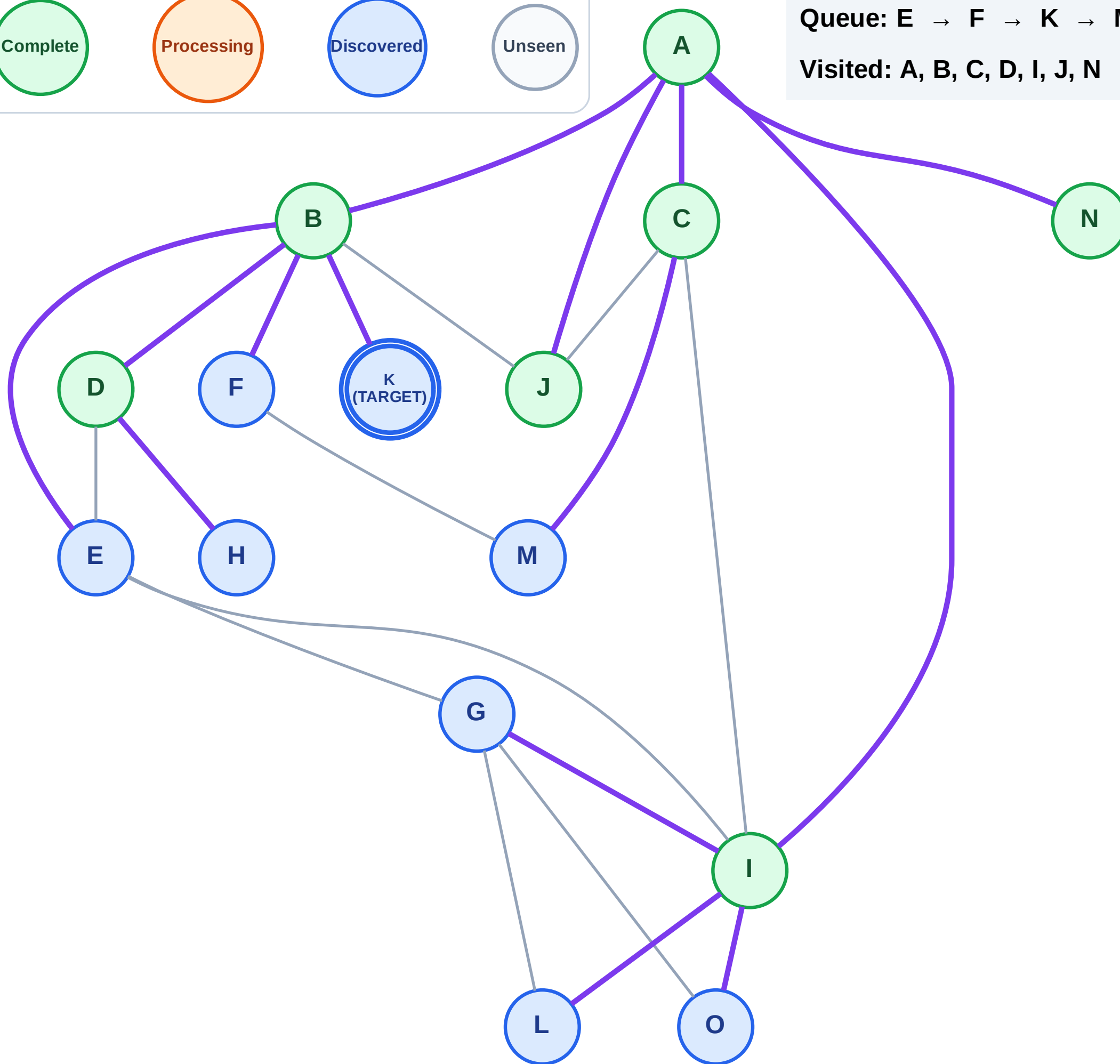
Discovered



Unseen

Queue: E → F → K → M → G → L → O → H

Visited: A, B, C, D, I, J, N



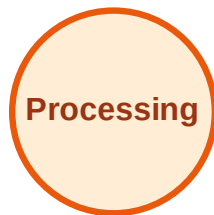
BFS Traversal

Step 16: Process node E

Legend



Complete



Processing



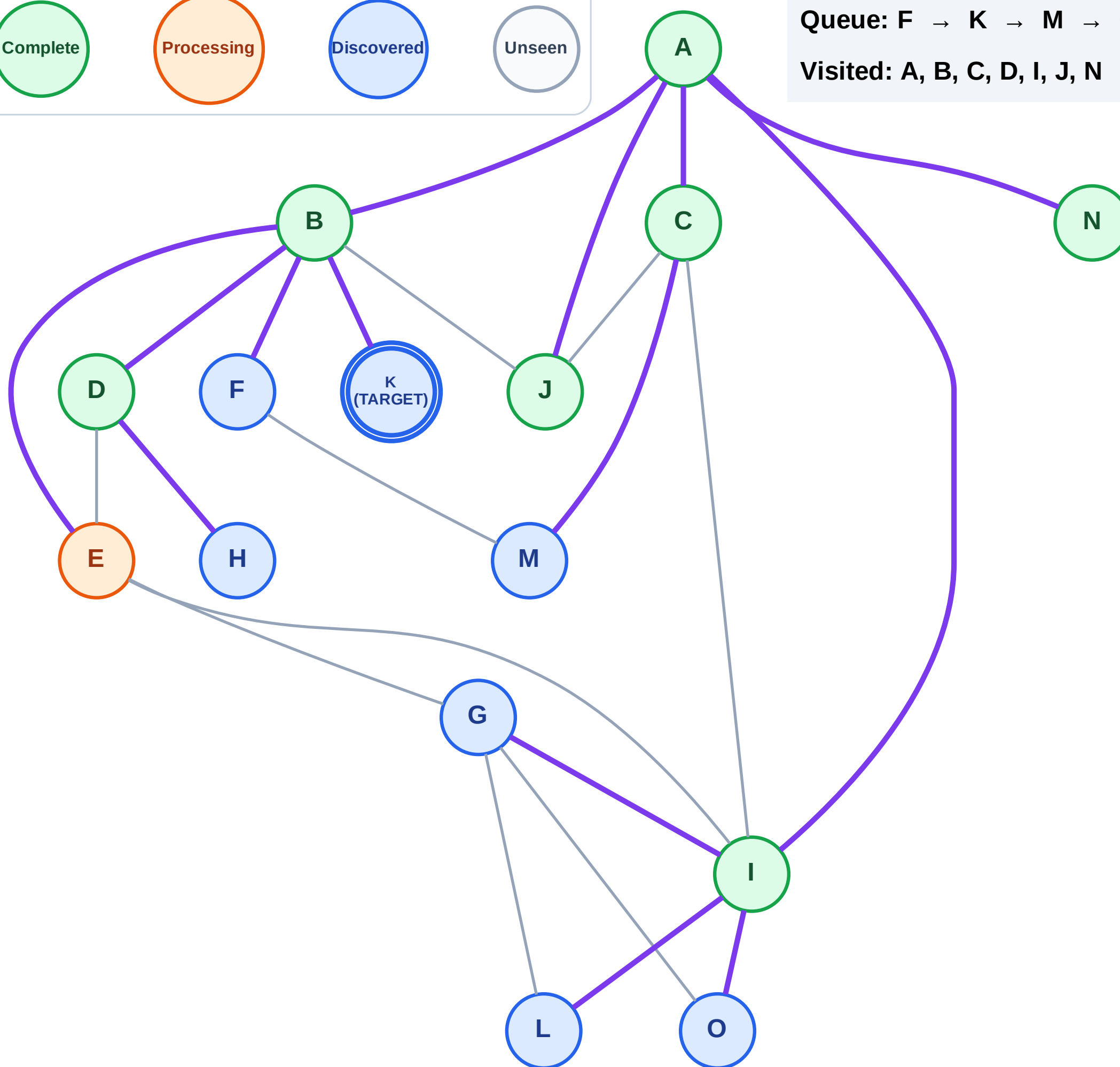
Discovered



Unseen

Queue: F → K → M → G → L → O → H

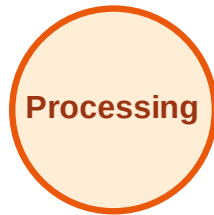
Visited: A, B, C, D, I, J, N



BFS Traversal

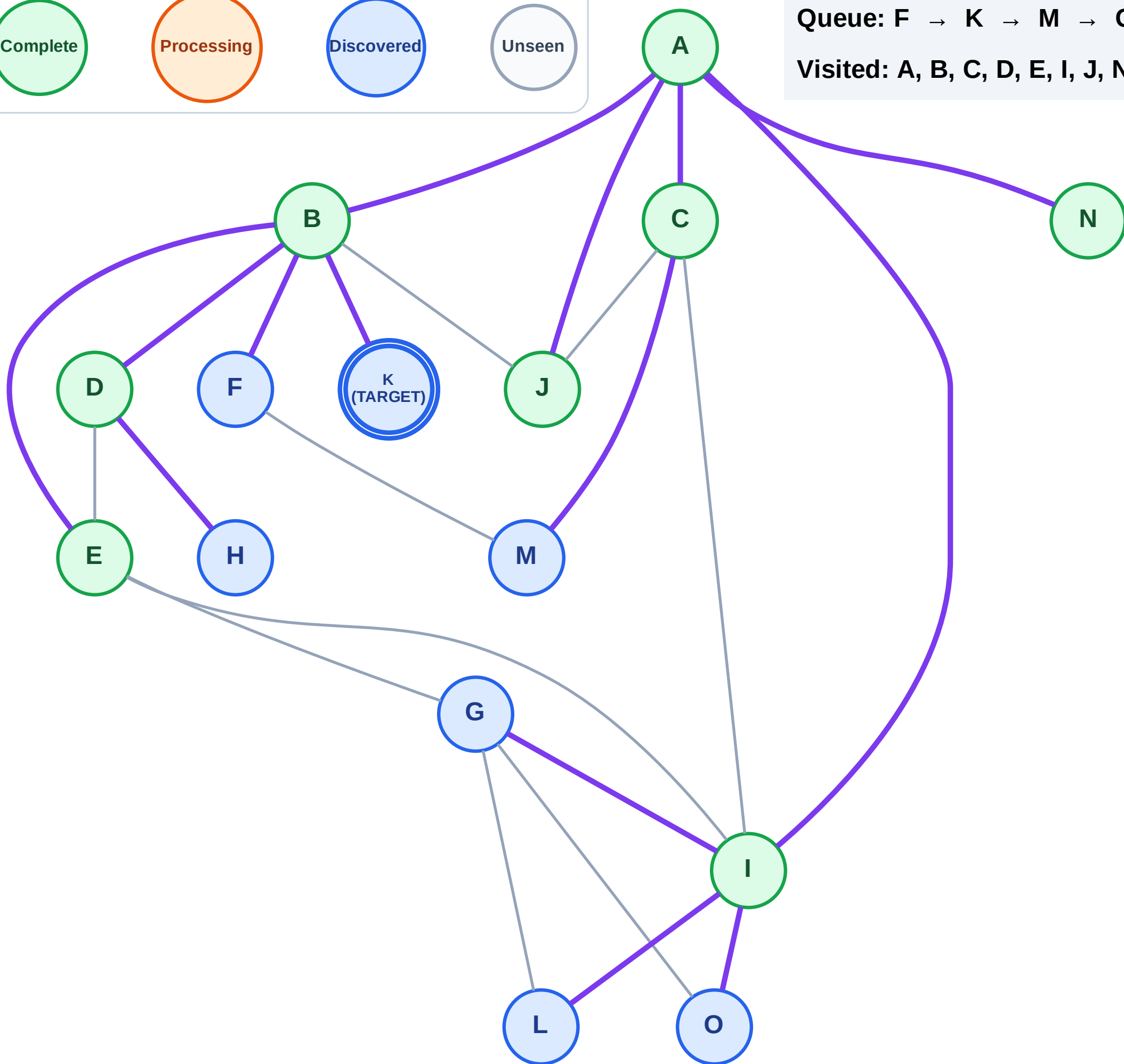
Step 17: No new neighbors from E

Legend



Queue: F → K → M → G → L → O → H

Visited: A, B, C, D, E, I, J, N



BFS Traversal

Step 18: Process node F

Legend

Complete

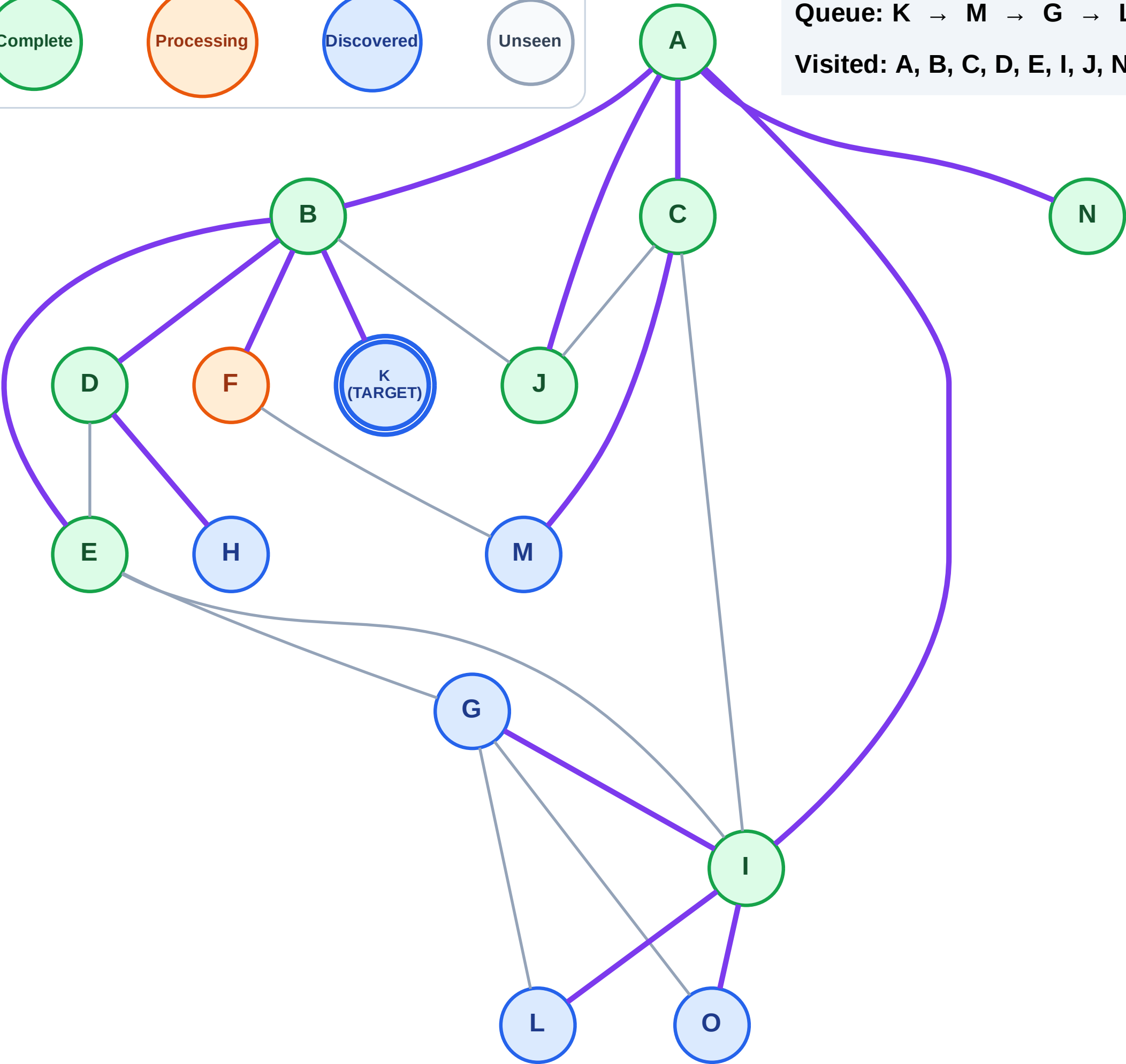
Processing

Discovered

Unseen

Queue: K → M → G → L → O → H

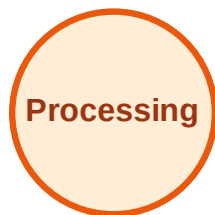
Visited: A, B, C, D, E, I, J, N



BFS Traversal

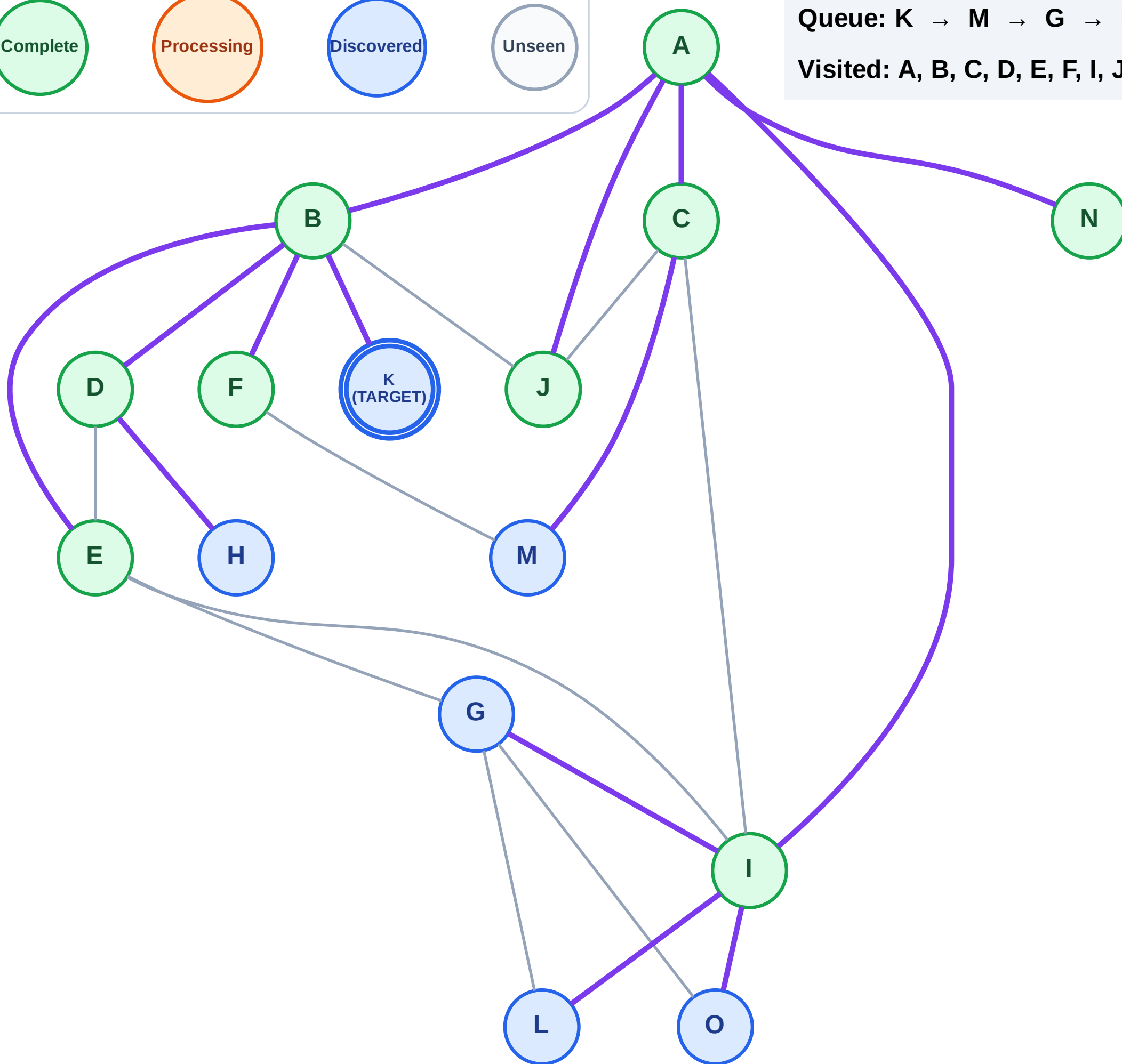
Step 19: No new neighbors from F

Legend



Queue: K → M → G → L → O → H

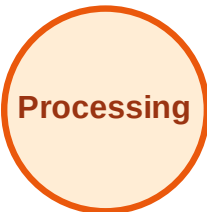
Visited: A, B, C, D, E, F, I, J, N



BFS Traversal

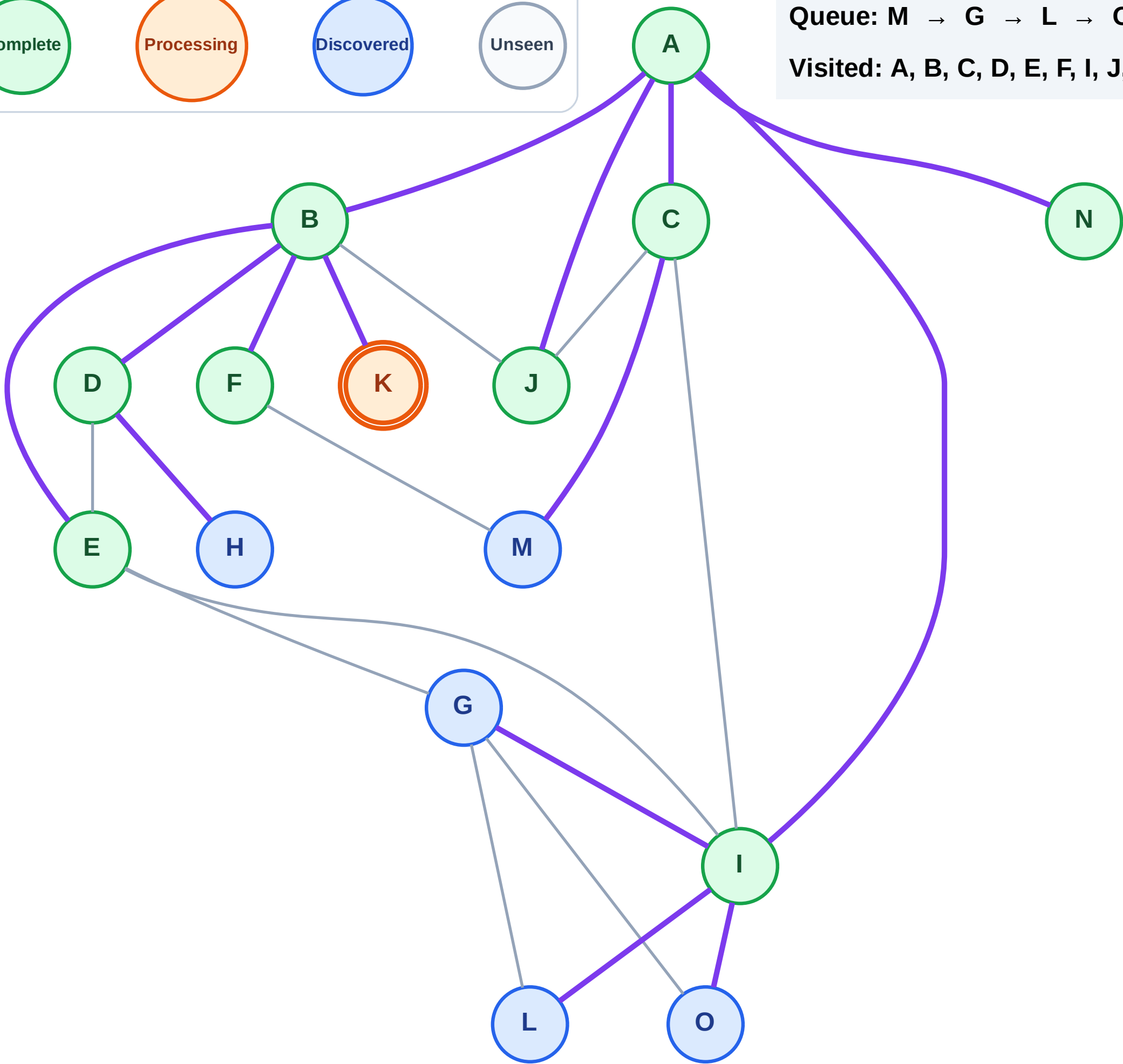
Step 20: Process node K

Legend



Queue: M → G → L → O → H

Visited: A, B, C, D, E, F, I, J, N



BFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

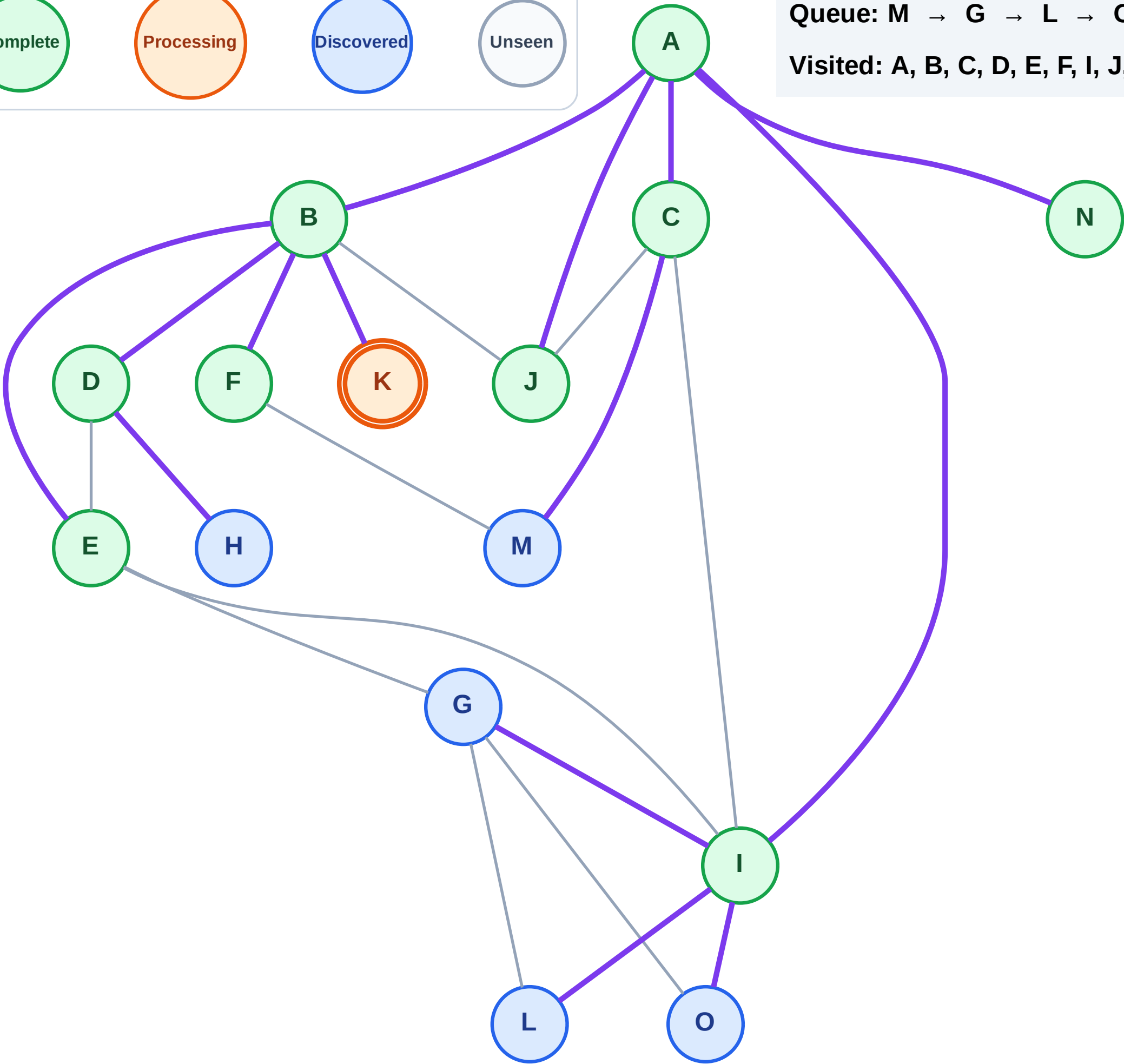
Processing

Discovered

Unseen

Queue: M → G → L → O → H

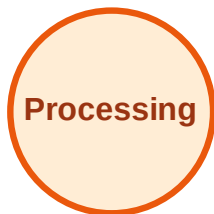
Visited: A, B, C, D, E, F, I, J, K, N



DFS Traversal

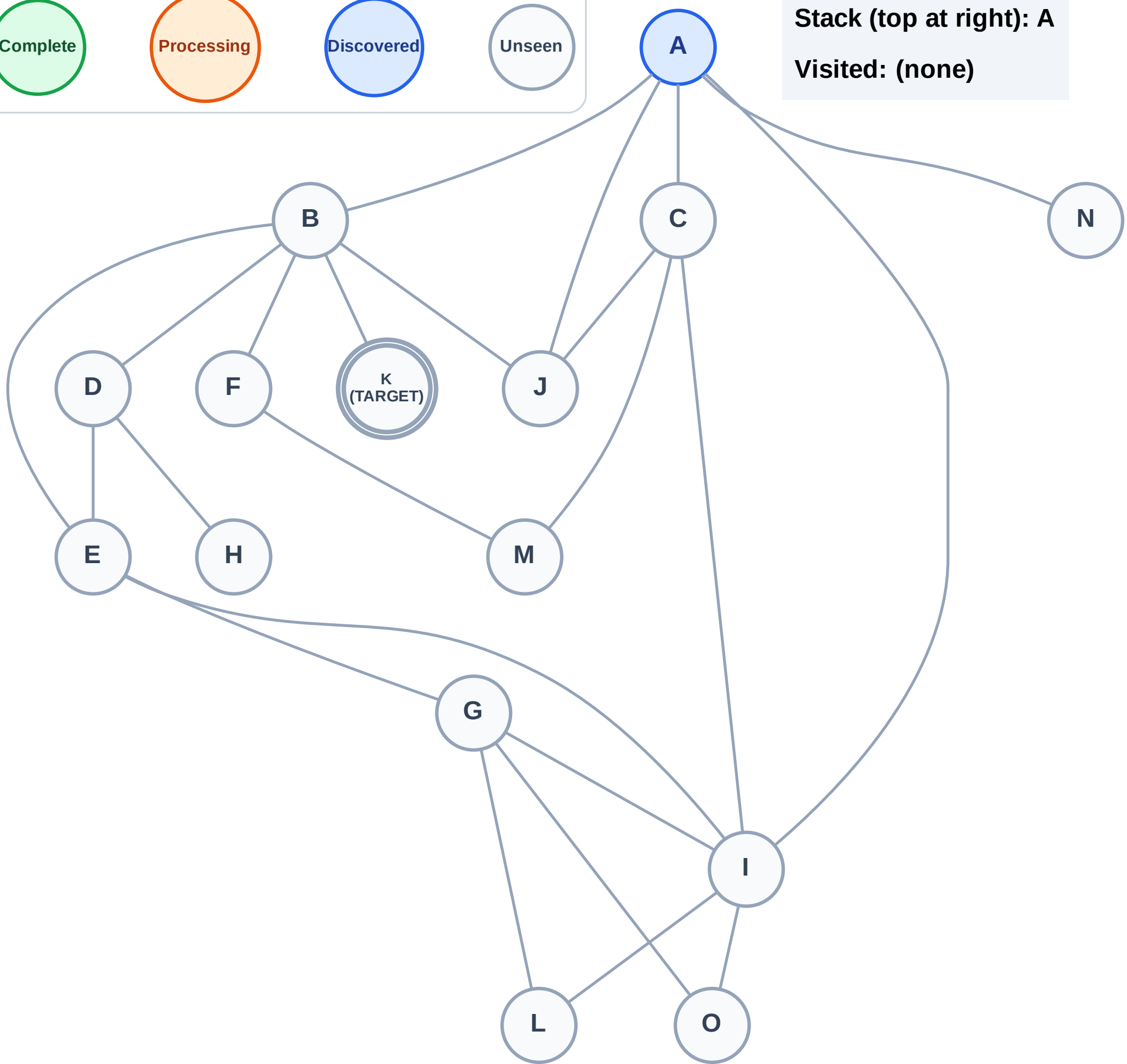
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

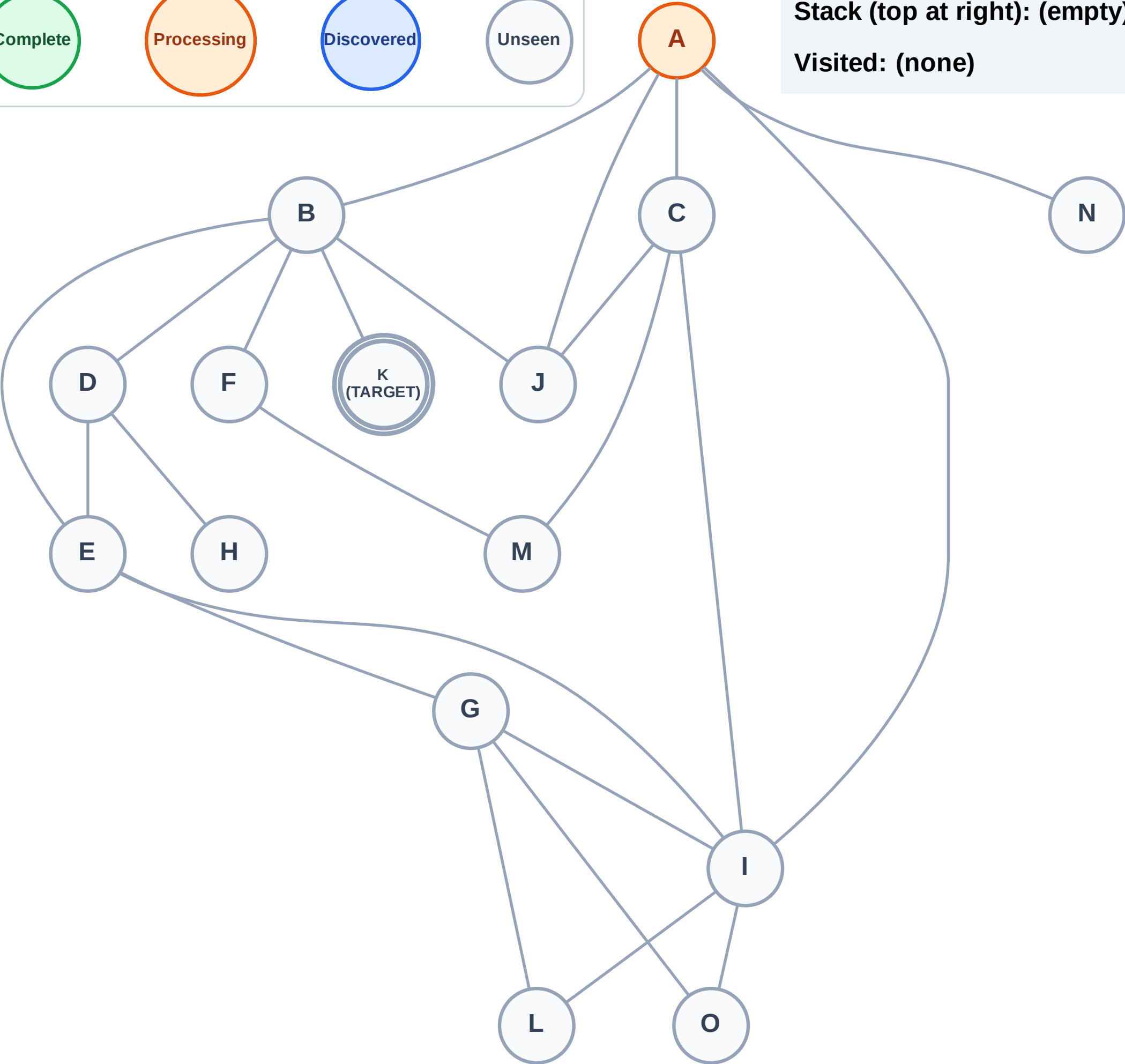
Complete

Processing

Discovered

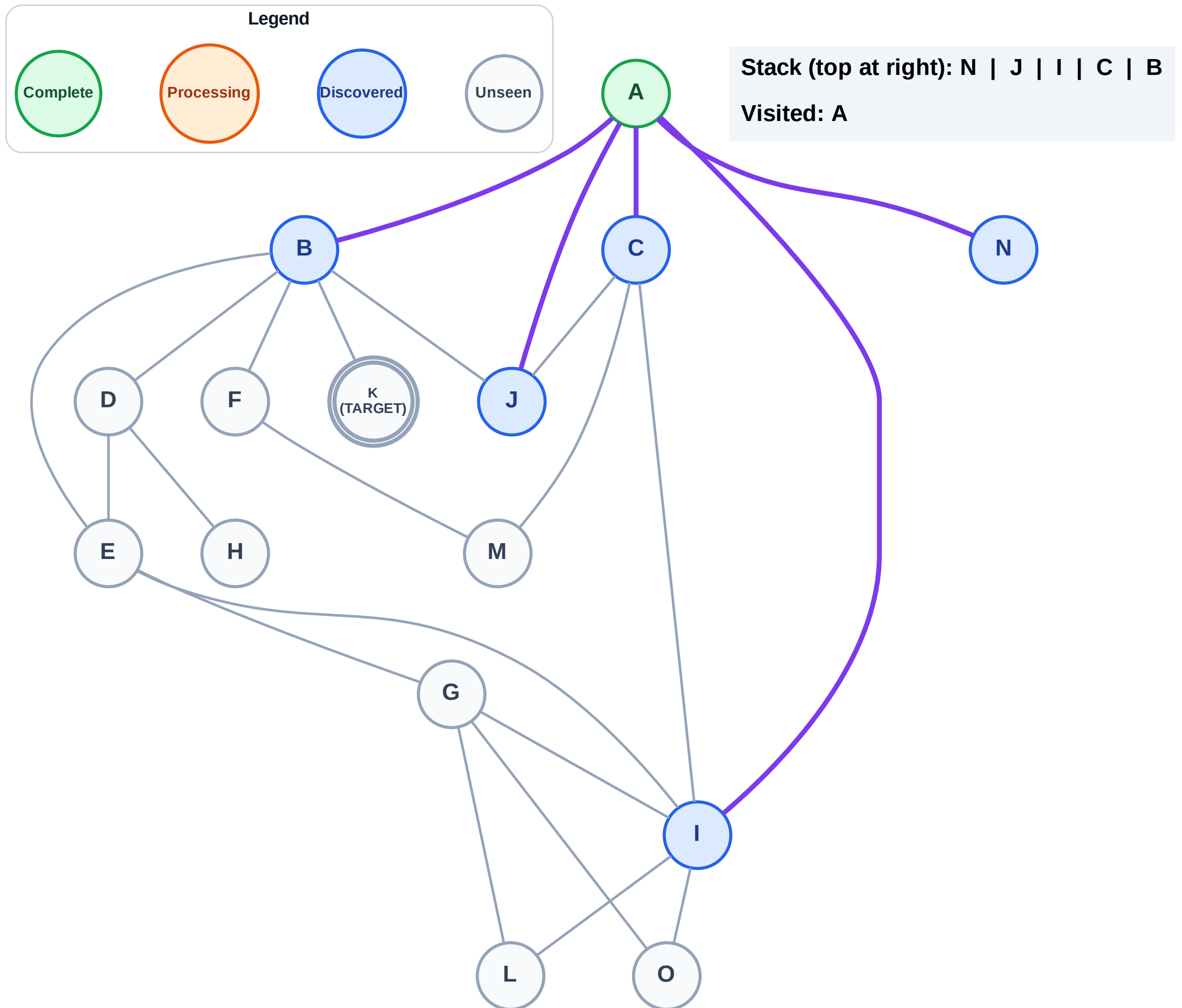
Unseen

Stack (top at right): (empty)
Visited: (none)



Step 3: Discover N, J, I, C, B from A

Step 3: Discover N, J, I, C, B from A



DFS Traversal

Step 4: Process node B

Legend

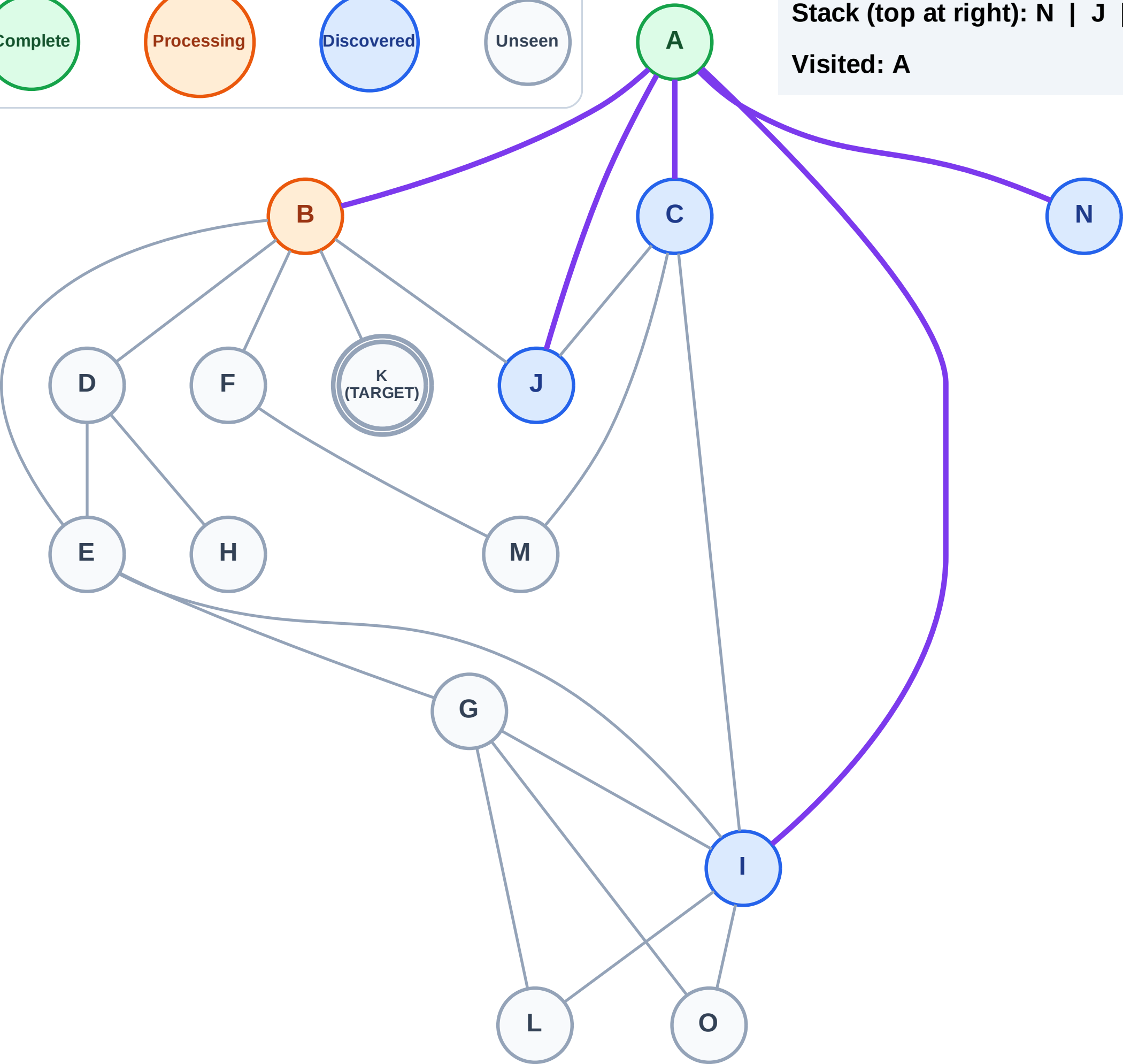
Complete

Processing

Discovered

Unseen

Stack (top at right): N | J | I | C
Visited: A



DFS Traversal

Step 5: Discover K, F, E, D from B

Legend

Complete

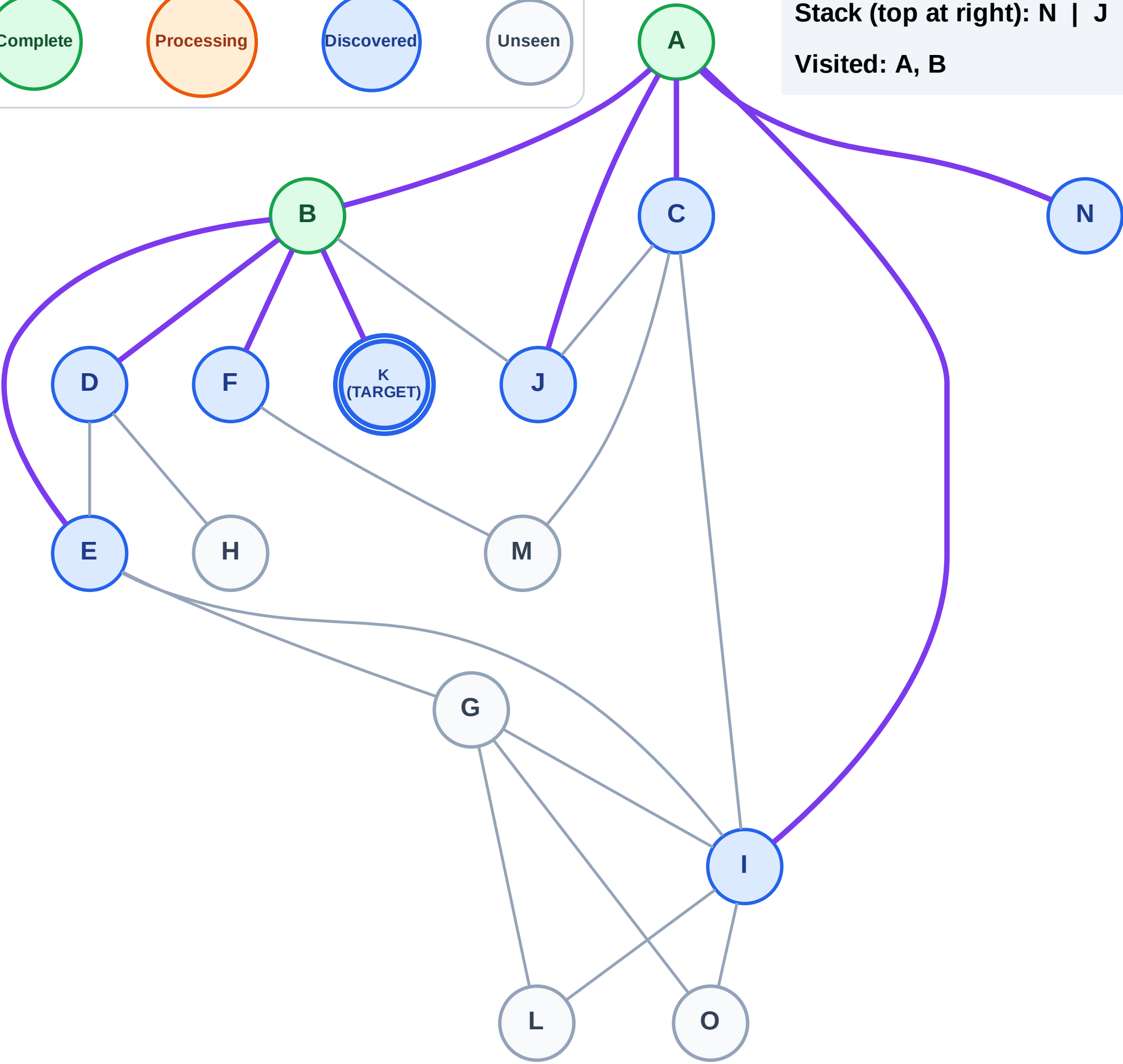
Processing

Discovered

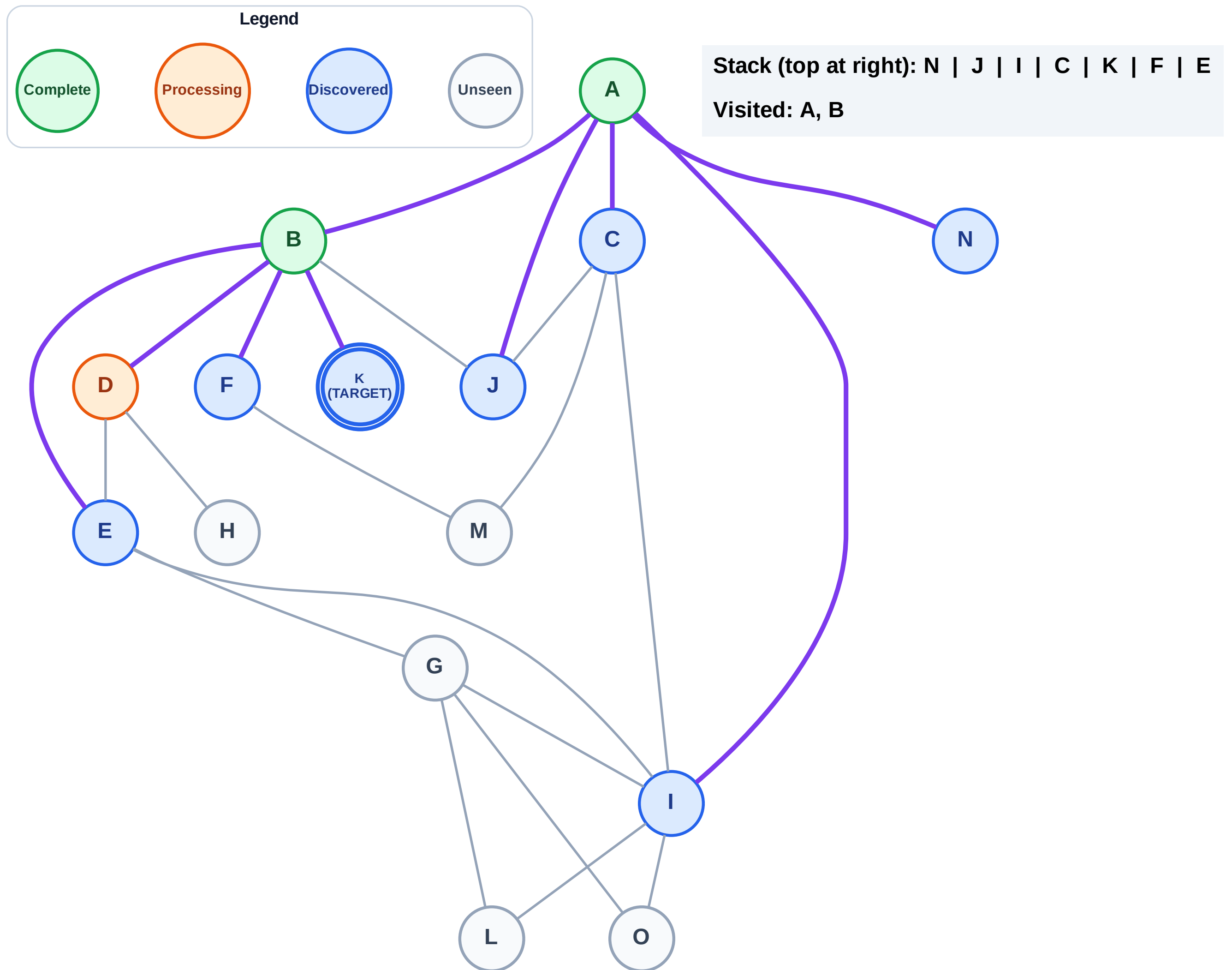
Unseen

Stack (top at right): N | J | I | C | K | F | E | D

Visited: A, B

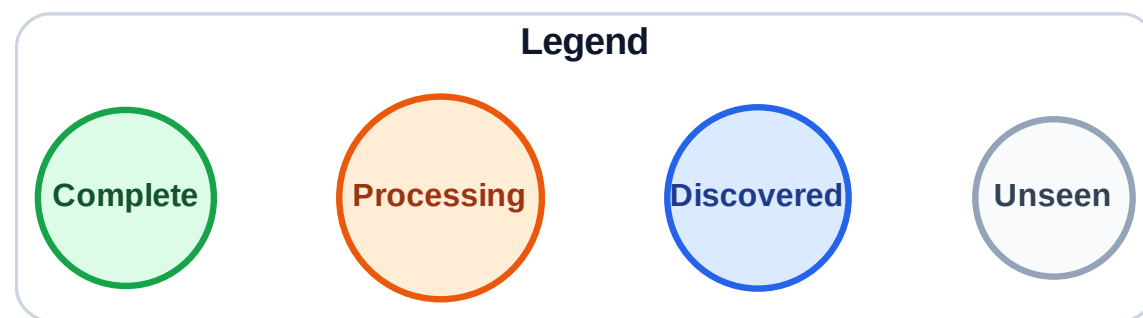


Step 6: Process node D

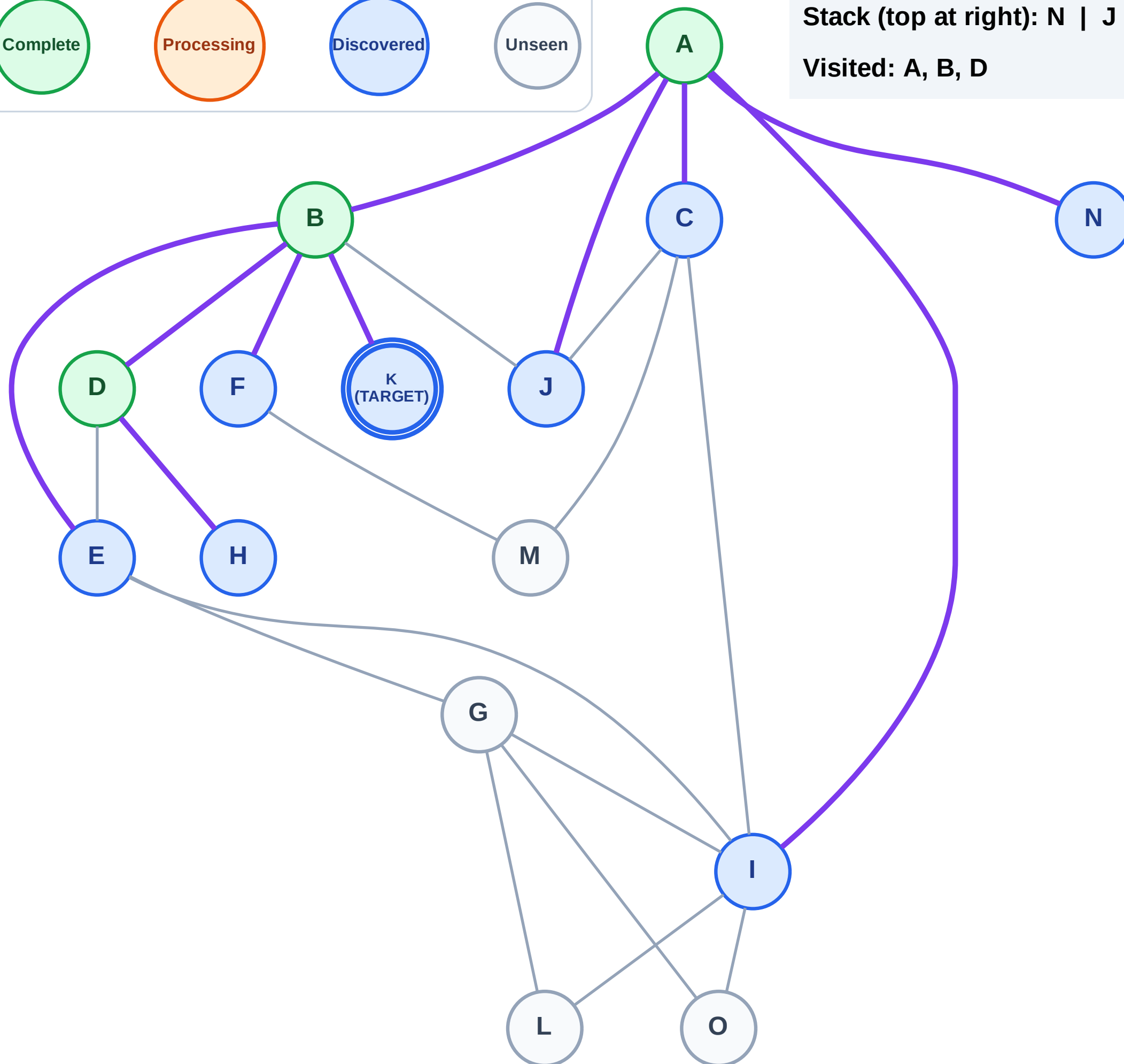


DFS Traversal

Step 7: Discover H from D



Stack (top at right): N | J | I | C | K | F | E | H
Visited: A, B, D



DFS Traversal

Step 8: Process node H

Legend

Complete

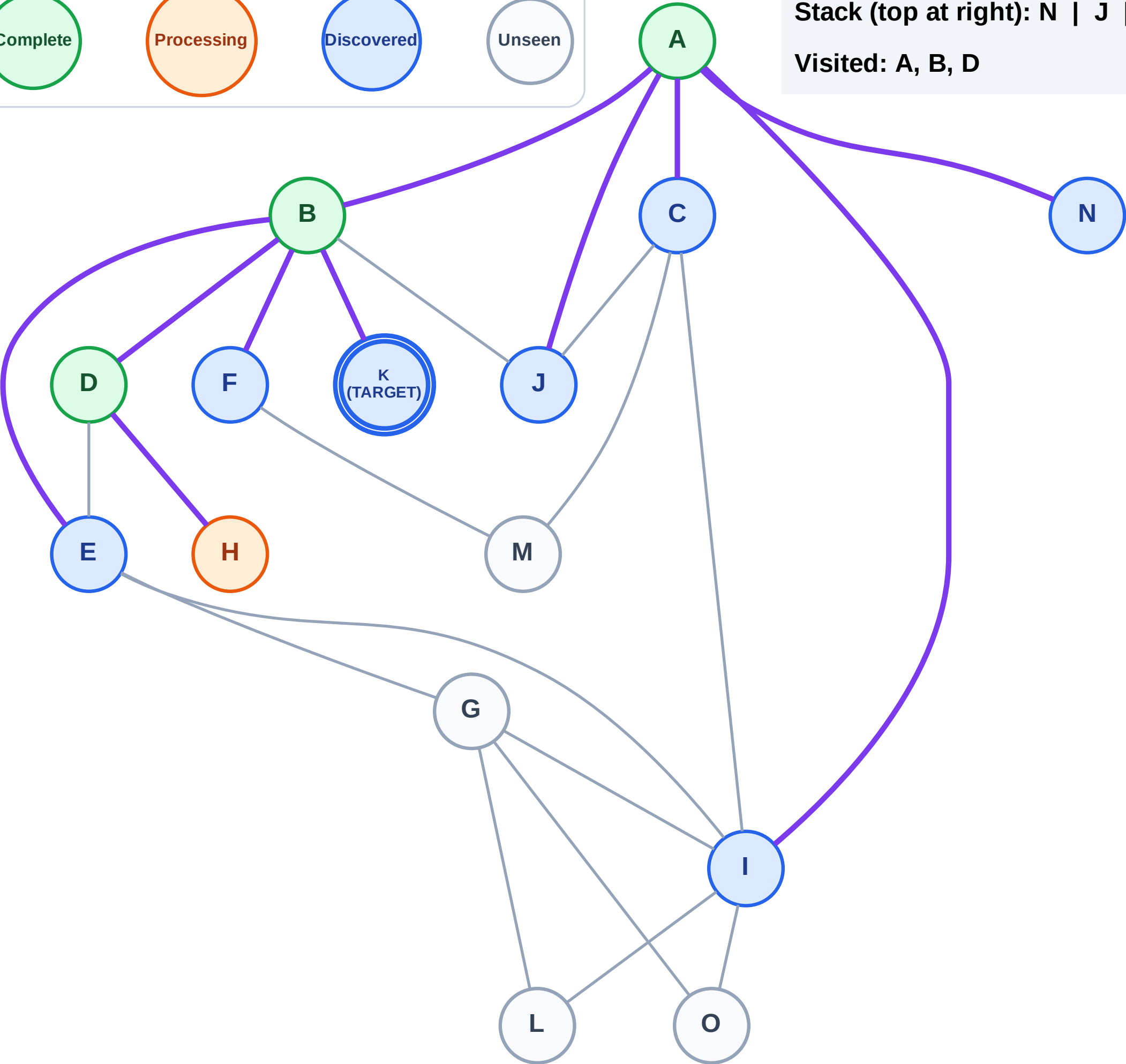
Processing

Discovered

Unseen

Stack (top at right): N | J | I | C | K | F | E

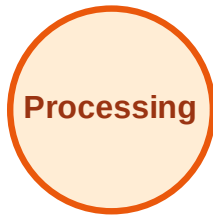
Visited: A, B, D



DFS Traversal

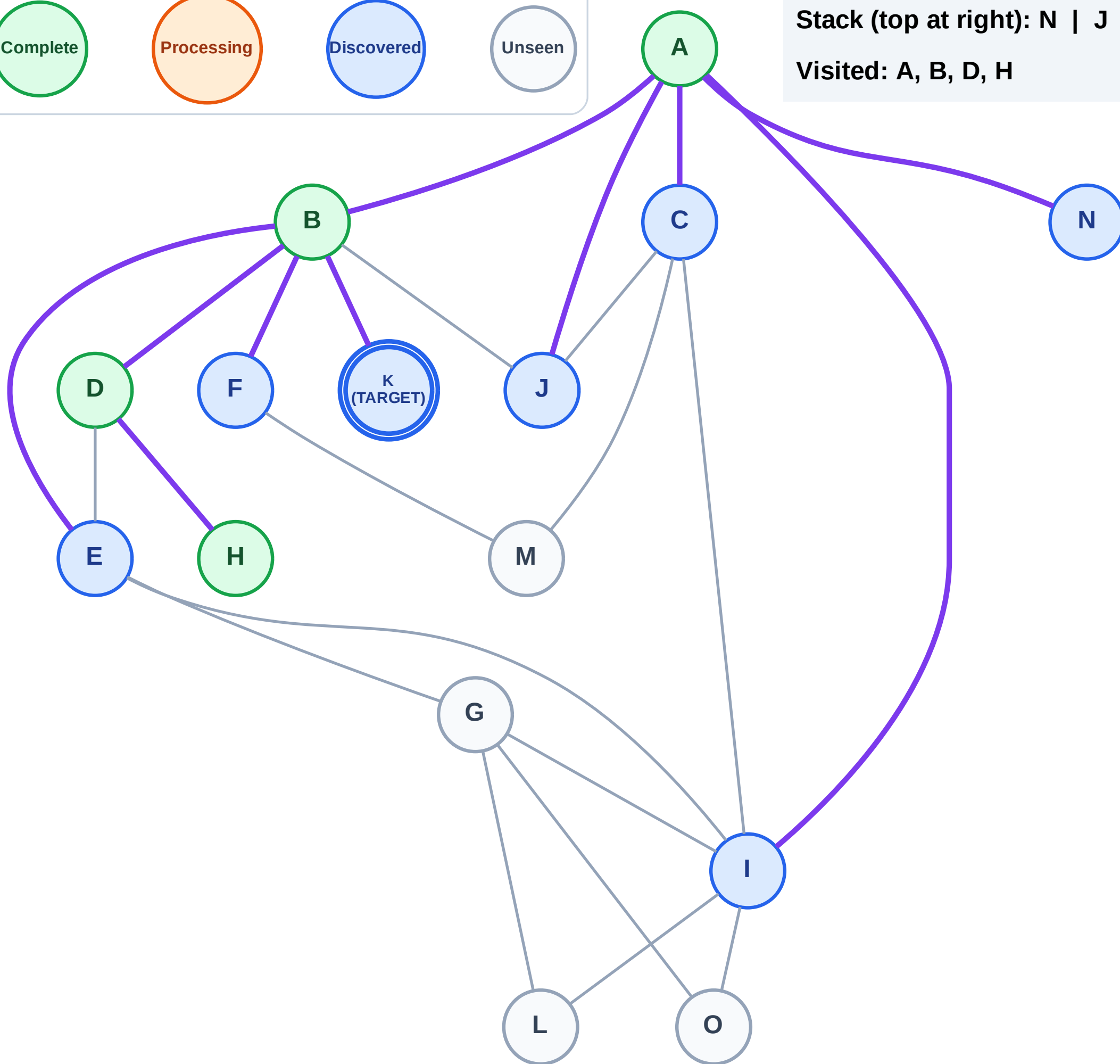
Step 9: No new neighbors from H

Legend



Stack (top at right): N | J | I | C | K | F | E

Visited: A, B, D, H



DFS Traversal

Step 10: Process node E

Legend

Complete

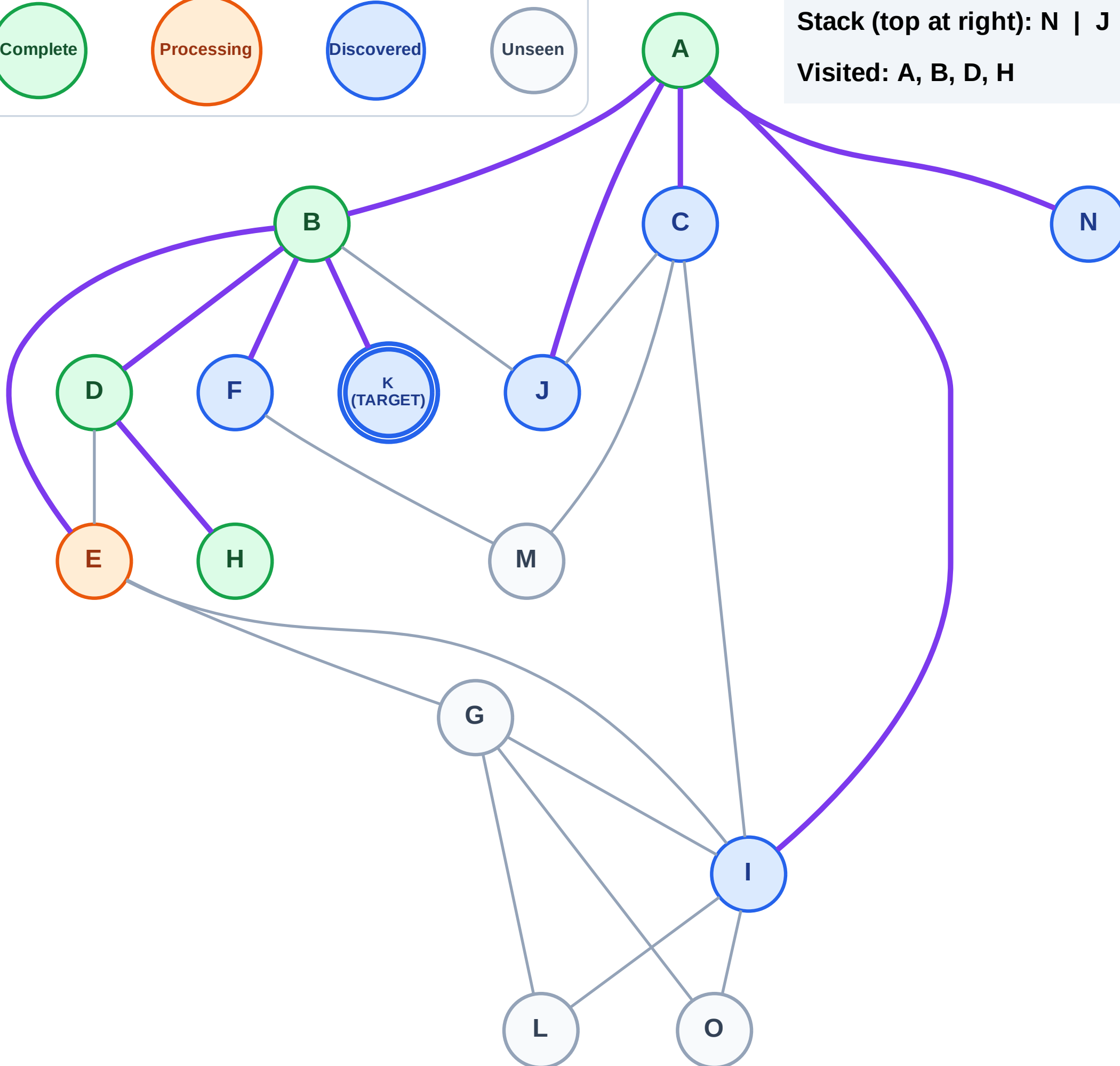
Processing

Discovered

Unseen

Stack (top at right): N | J | I | C | K | F

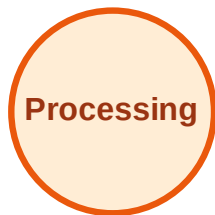
Visited: A, B, D, H



DFS Traversal

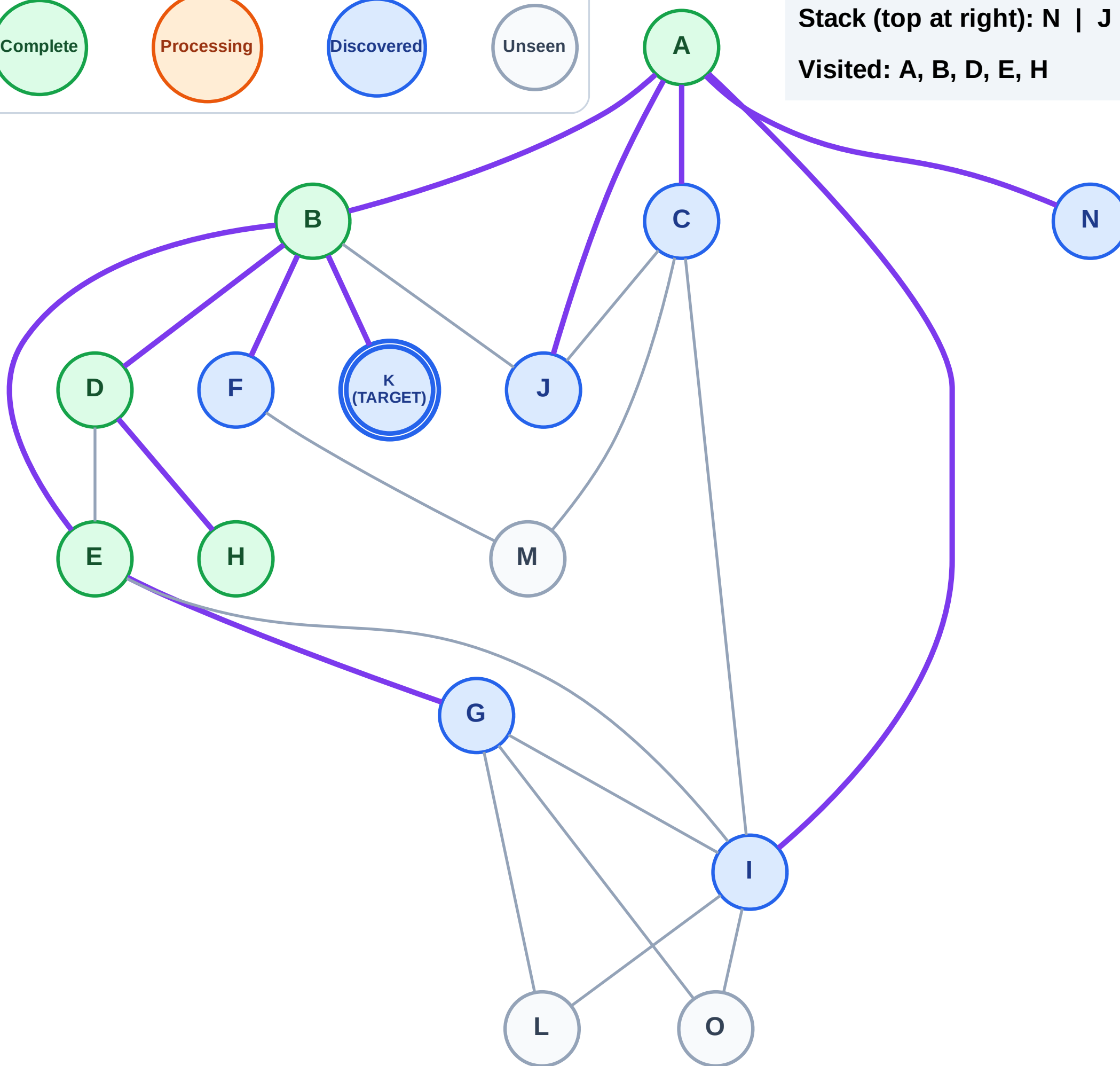
Step 11: Discover G from E

Legend



Stack (top at right): N | J | I | C | K | F | G

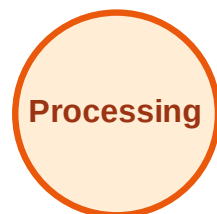
Visited: A, B, D, E, H



DFS Traversal

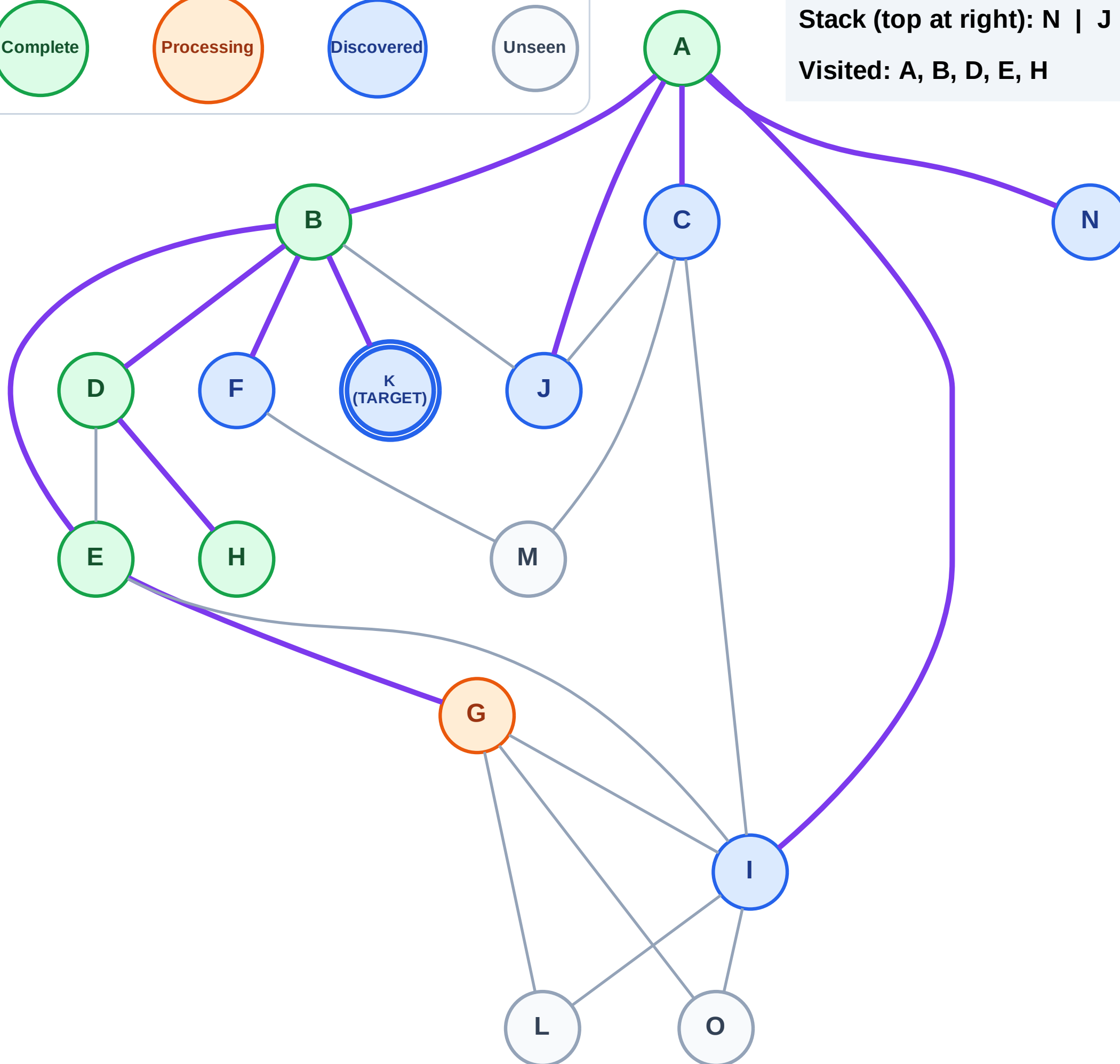
Step 12: Process node G

Legend



Stack (top at right): N | J | I | C | K | F

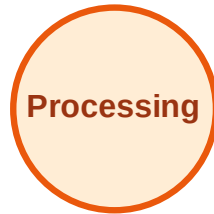
Visited: A, B, D, E, H



DFS Traversal

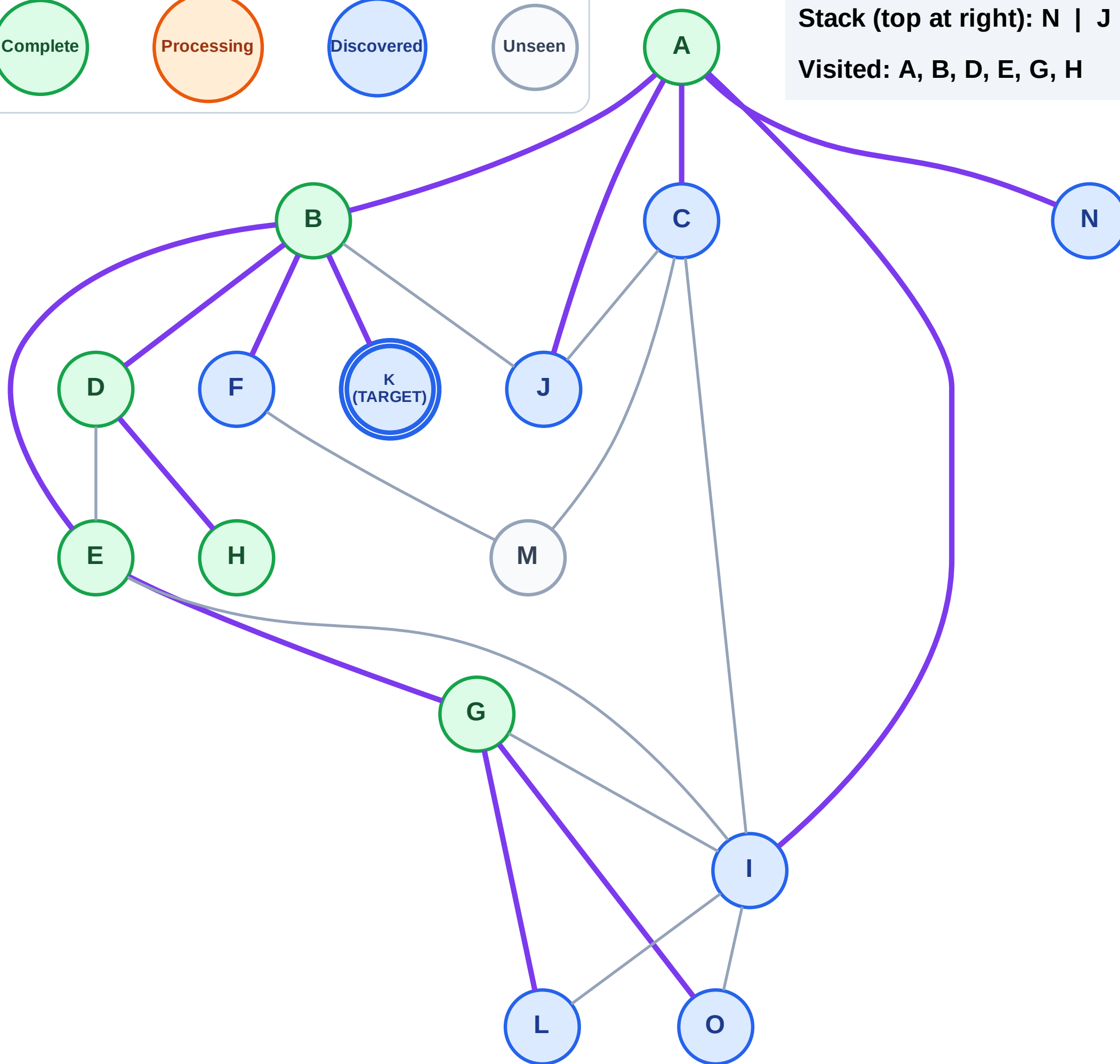
Step 13: Discover O, L from G

Legend

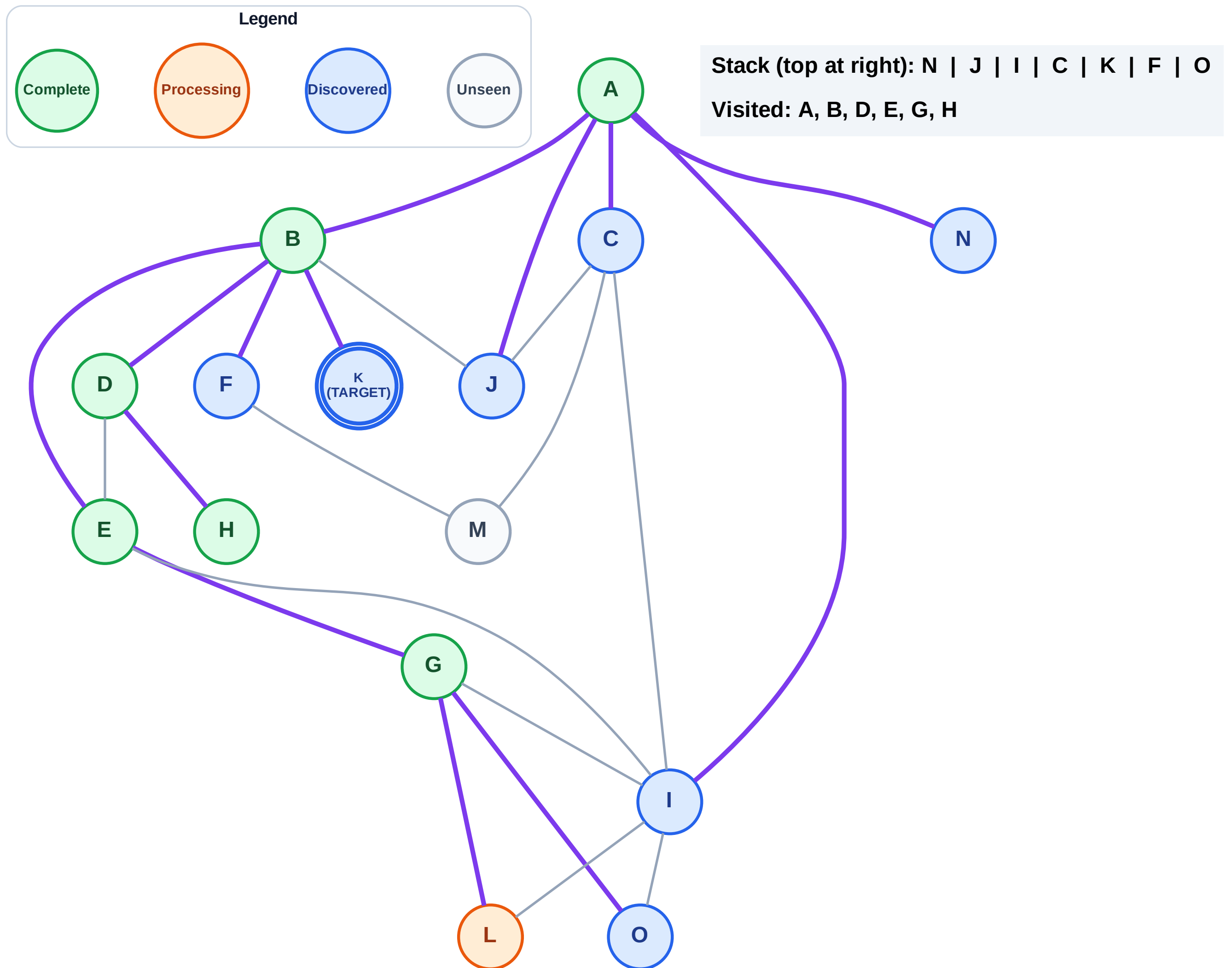


Stack (top at right): N | J | I | C | K | F | O | L

Visited: A, B, D, E, G, H



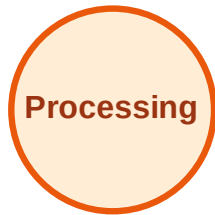
Step 14: Process node L



DFS Traversal

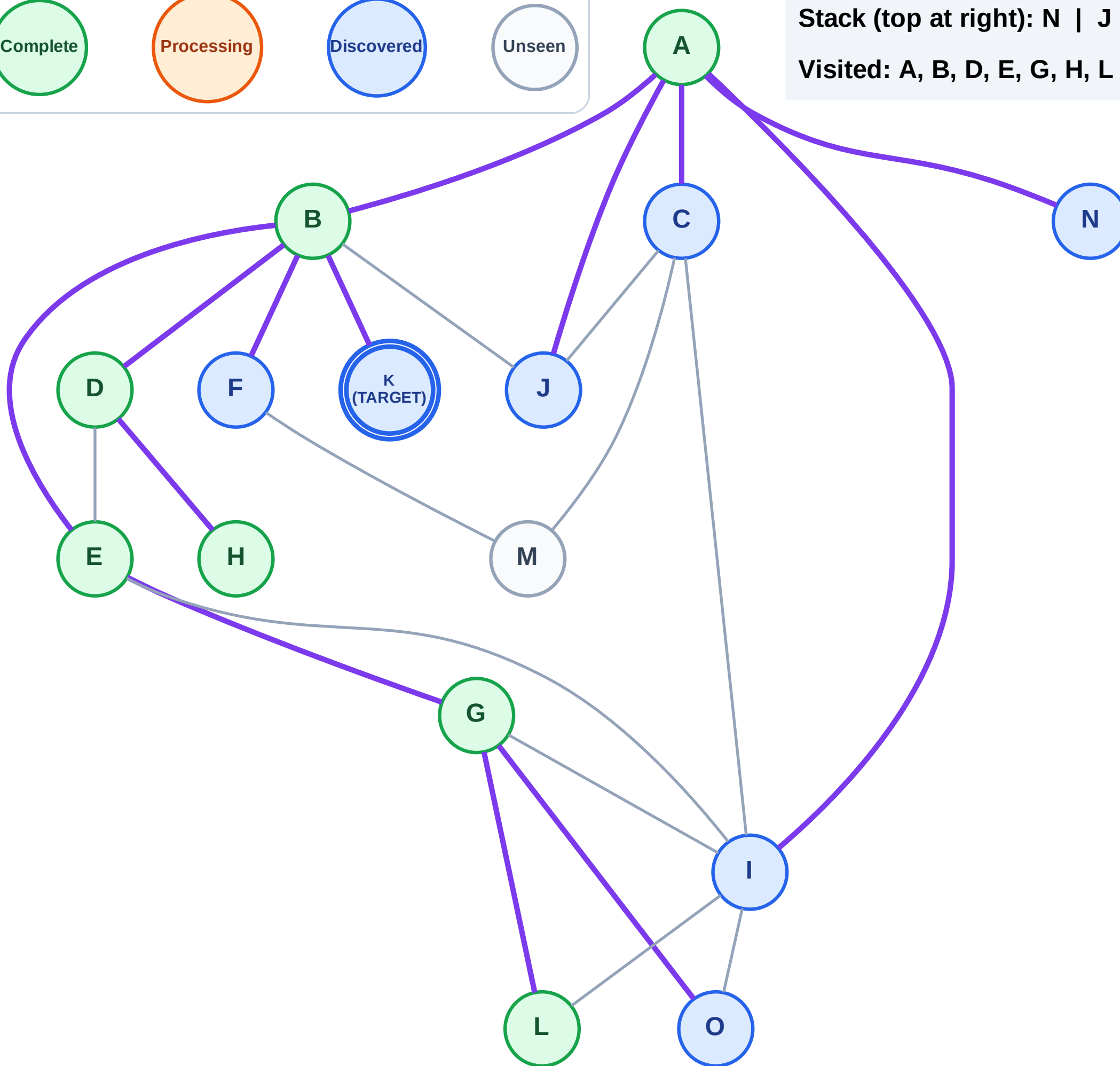
Step 15: No new neighbors from L

Legend



Stack (top at right): N | J | I | C | K | F | O

Visited: A, B, D, E, G, H, L



DFS Traversal

Step 16: Process node O

Legend

Complete

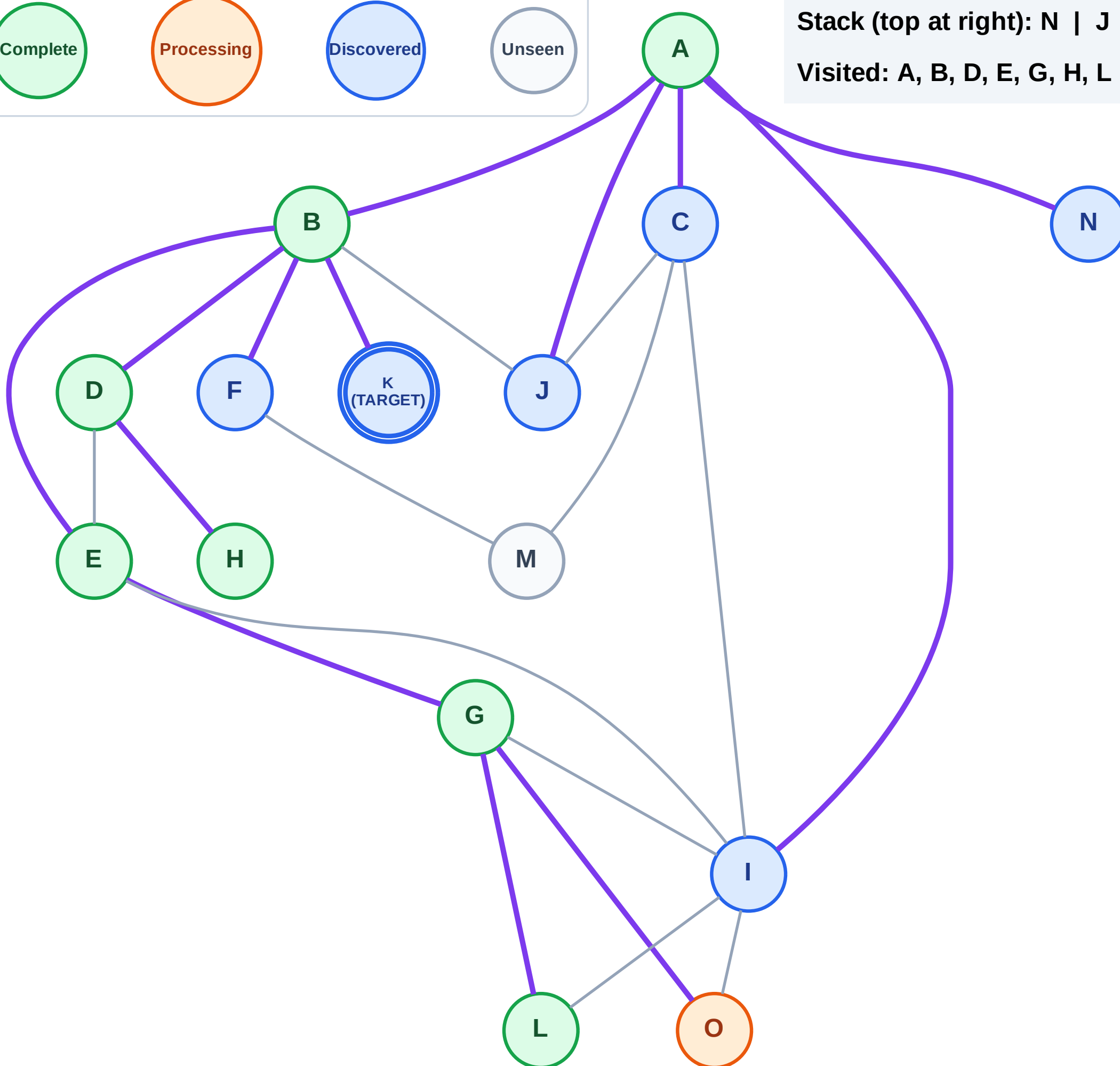
Processing

Discovered

Unseen

Stack (top at right): N | J | I | C | K | F

Visited: A, B, D, E, G, H, L



DFS Traversal

Step 17: No new neighbors from O

Legend

Complete

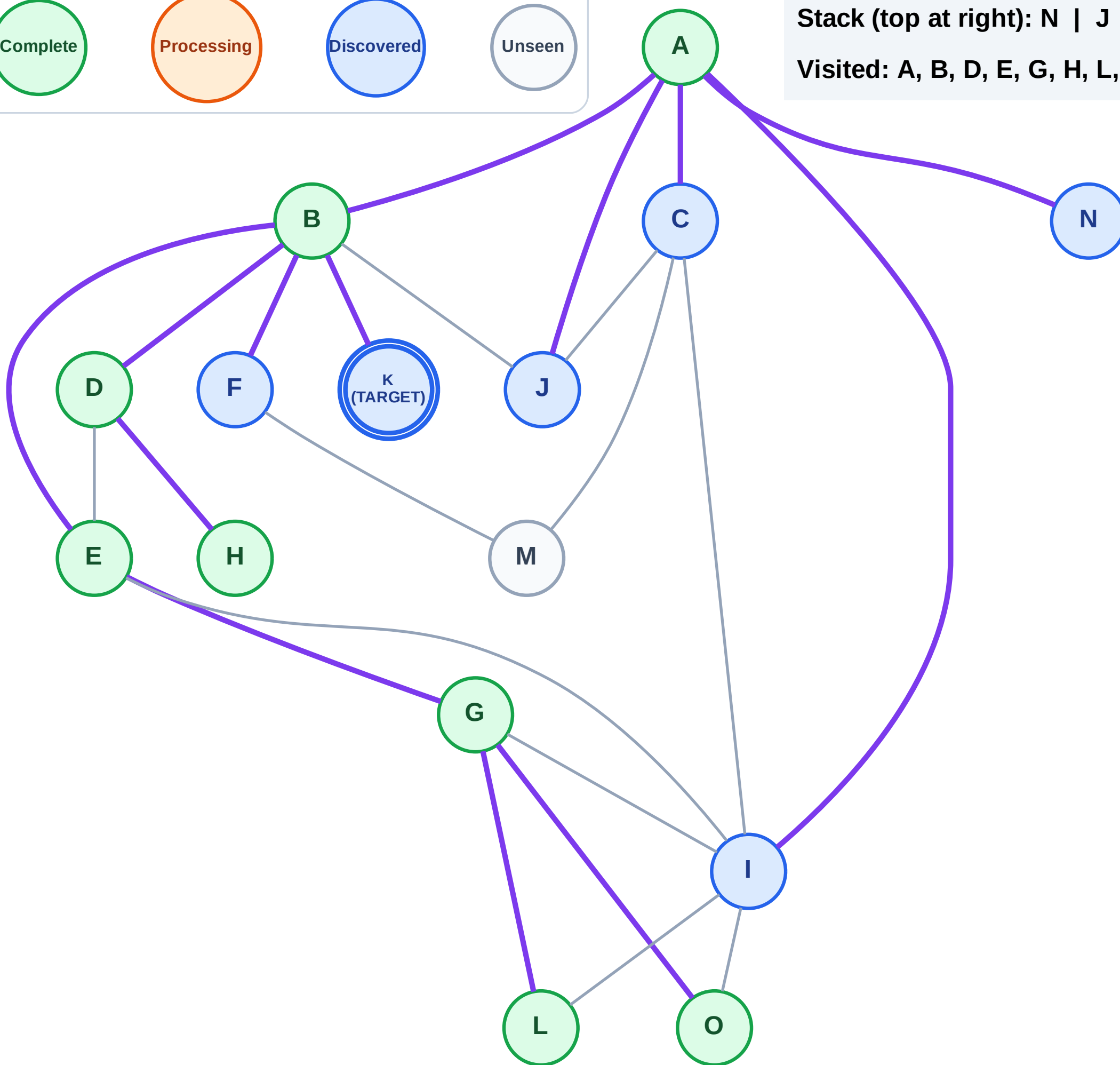
Processing

Discovered

Unseen

Stack (top at right): N | J | I | C | K | F

Visited: A, B, D, E, G, H, L, O



DFS Traversal

Step 18: Process node F

Stack (top at right): N | J | I | C | K
Visited: A, B, D, E, G, H, L, O

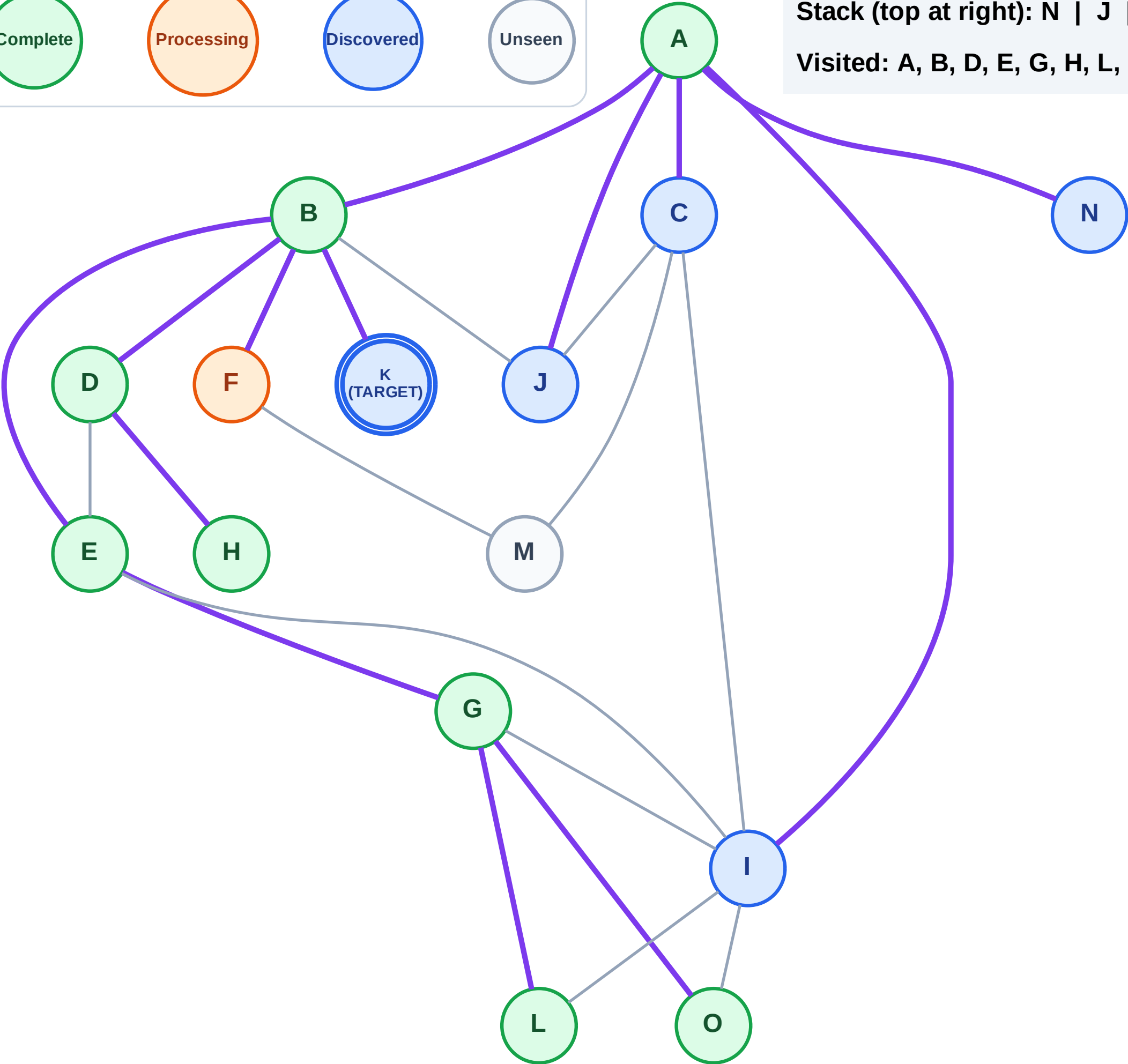
Legend

Complete

Processing

Discovered

Unseen



DFS Traversal

Step 19: Discover M from F

Legend

Complete

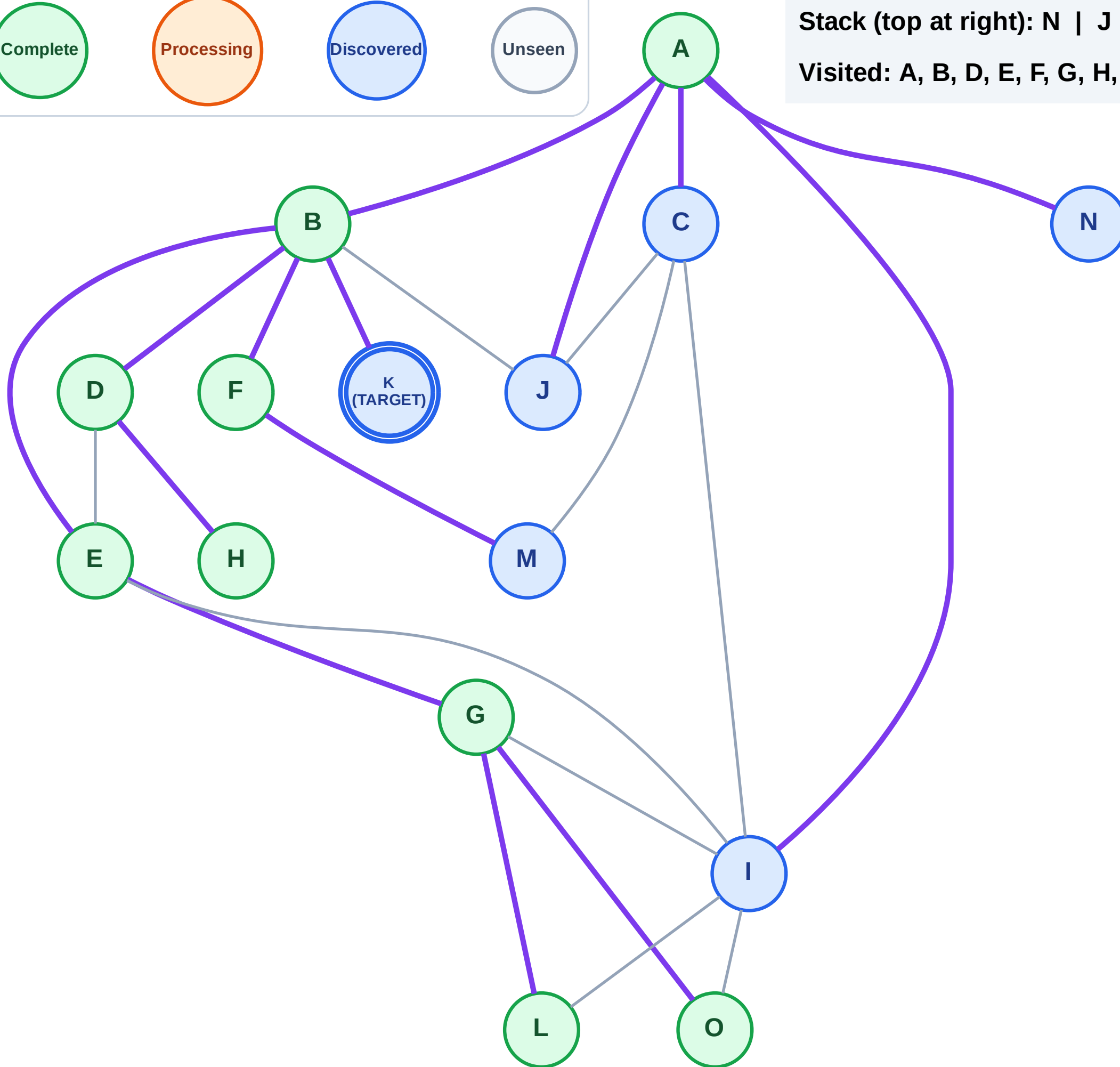
Processing

Discovered

Unseen

Stack (top at right): N | J | I | C | K | M

Visited: A, B, D, E, F, G, H, L, O



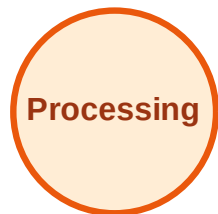
DFS Traversal

Step 20: Process node M

Legend



Complete



Processing



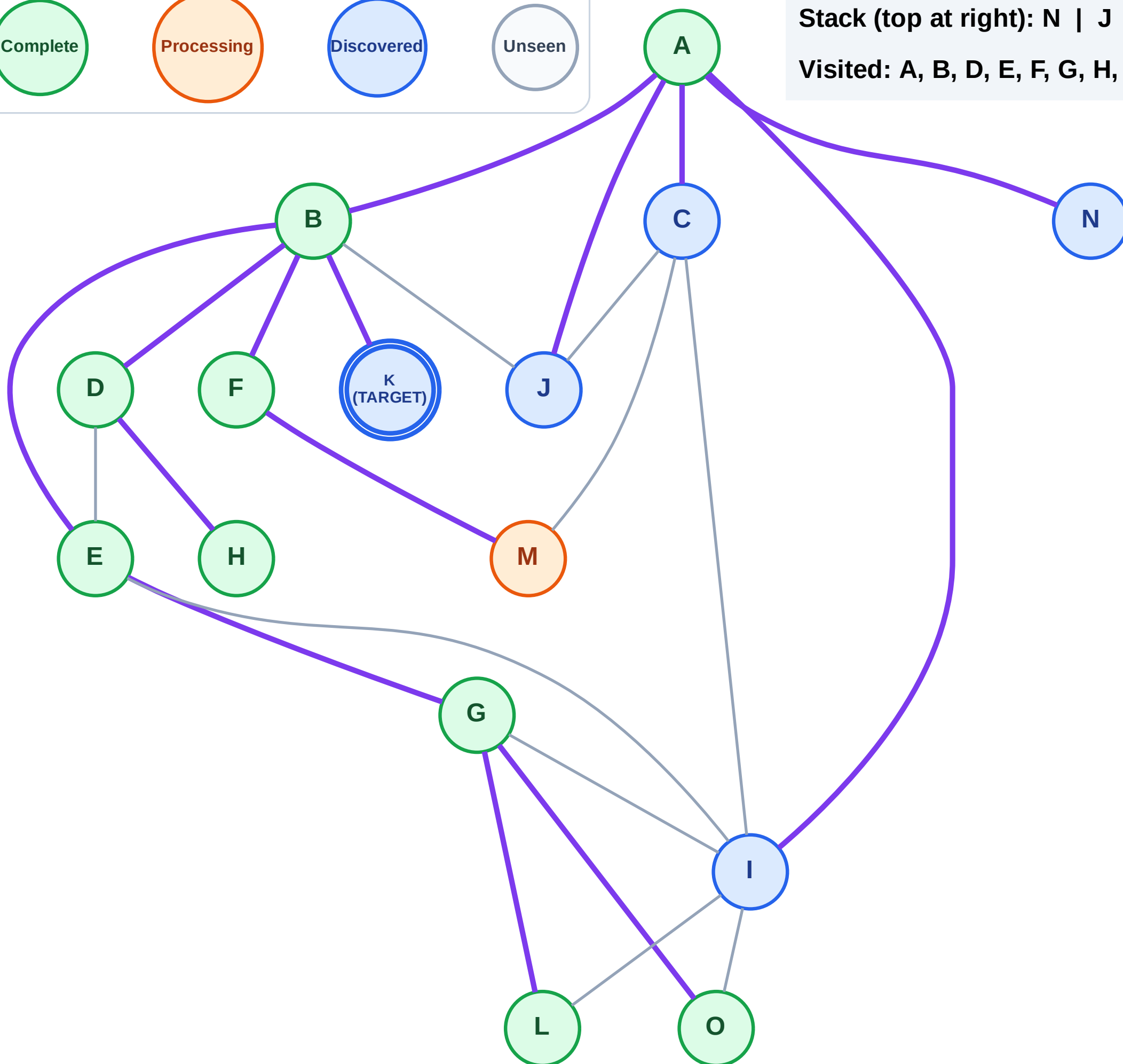
Discovered



Unseen

Stack (top at right): N | J | I | C | K

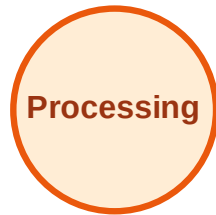
Visited: A, B, D, E, F, G, H, L, O



DFS Traversal

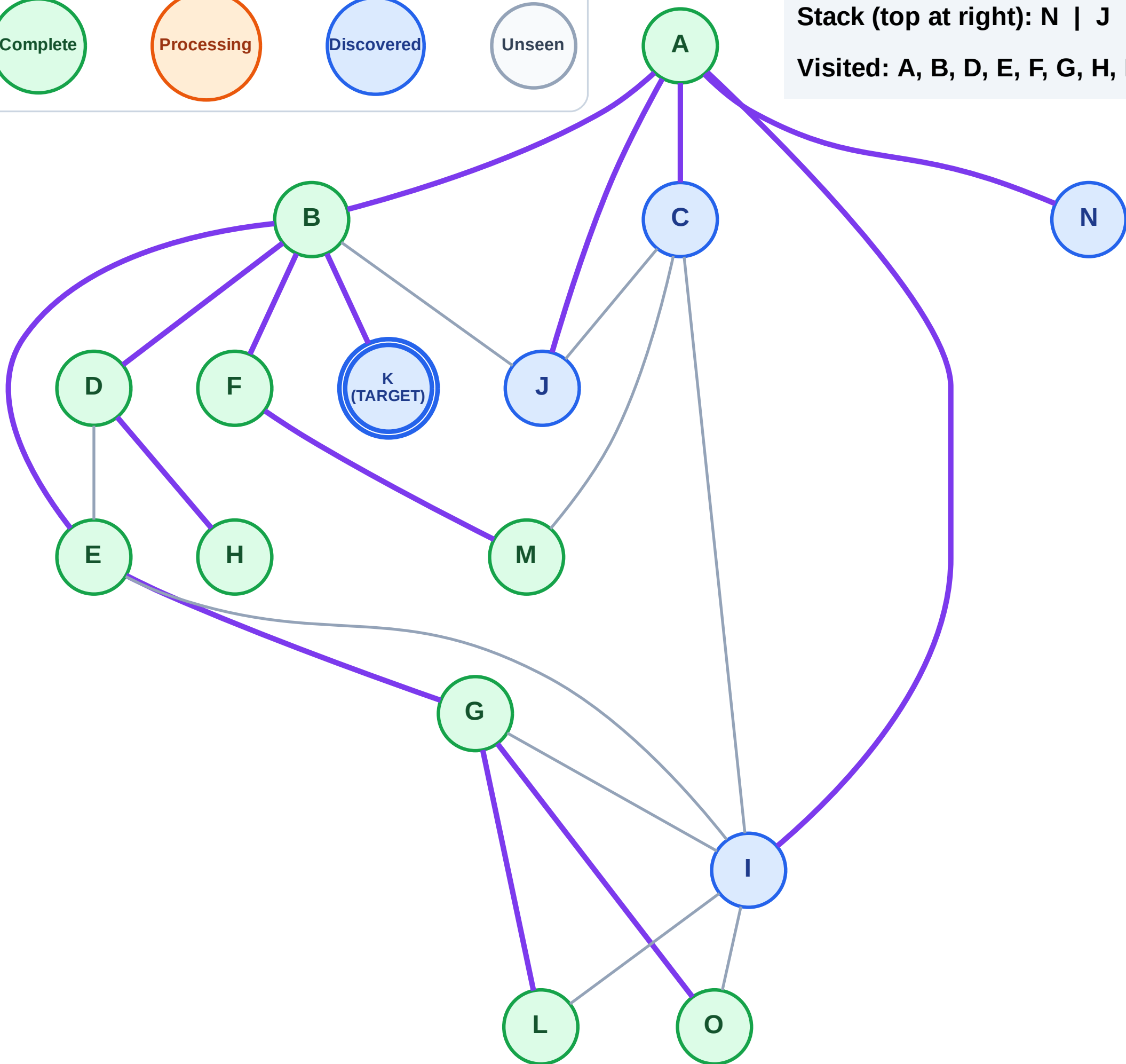
Step 21: No new neighbors from M

Legend



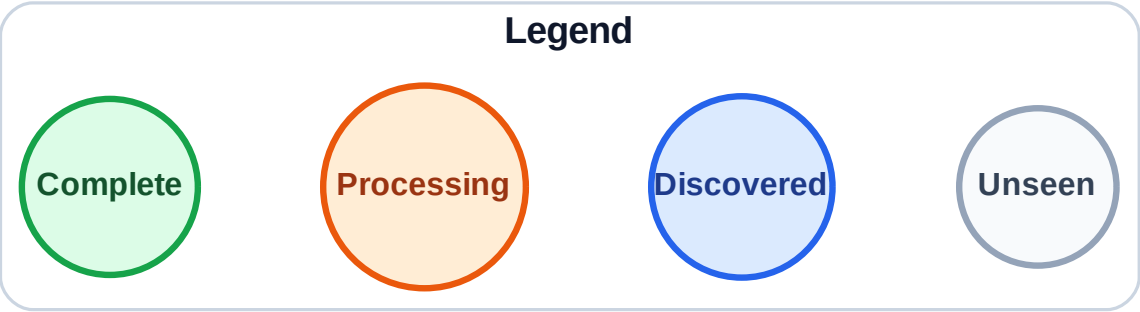
Stack (top at right): N | J | I | C | K

Visited: A, B, D, E, F, G, H, L, M, O



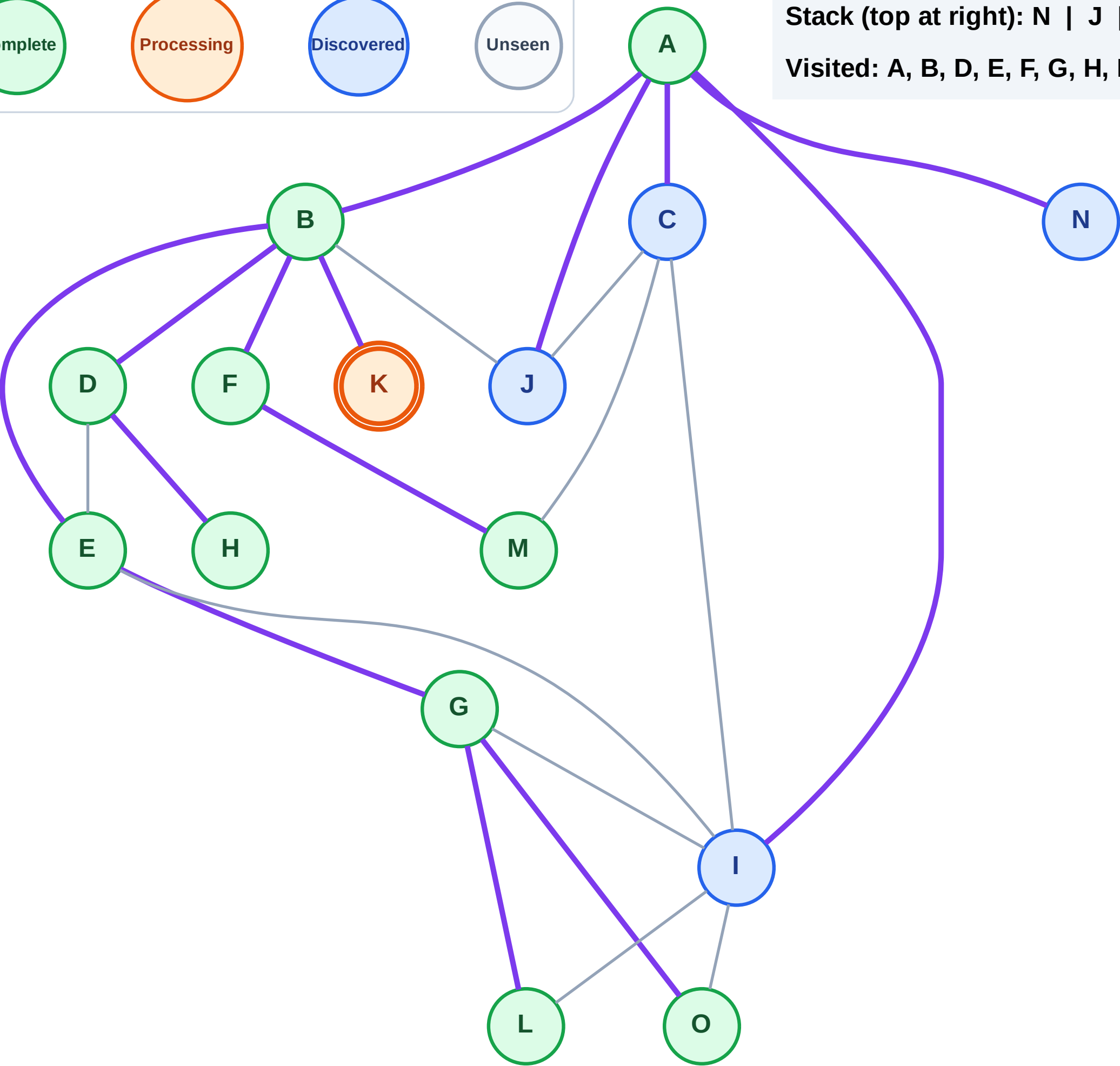
DFS Traversal

Step 22: Process node K



Stack (top at right): N | J | I | C

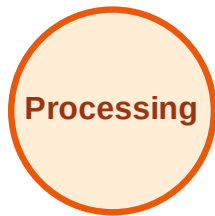
Visited: A, B, D, E, F, G, H, L, M, O



DFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Stack (top at right): N | J | I | C

Visited: A, B, D, E, F, G, H, K, L, M, O

